Engineering Mechanics: Statics

14th Edition
SI Units

Chapter 7: Internal Forces

Chapter Objectives

- To use the method of sections to determine the internal loadings in a member at a specific point
- To show how to obtain the internal shear and moment throughout a member and express the result graphically in the form of shear and moment diagrams
- To analyze the forces and the shape of cables supporting various types of loadings

Chapter Outline

7.1 **Internal Forces Developed in Structural Members**

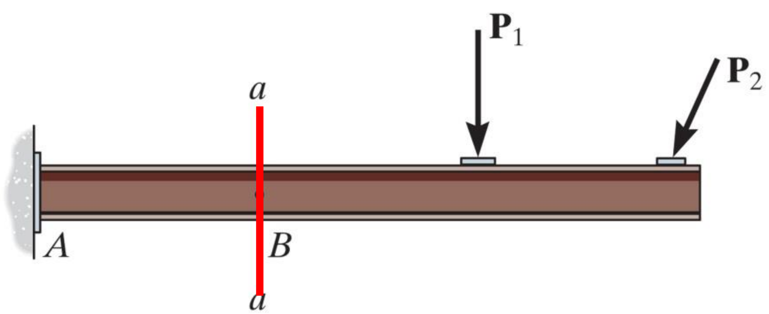
7.2 **Shear and Moment Equations and Diagrams**

7.3 **Relations between Distributed Load, Shear and Moment**

7.4 **Cables**

7.1 Internal Forces Developed in Structural Members

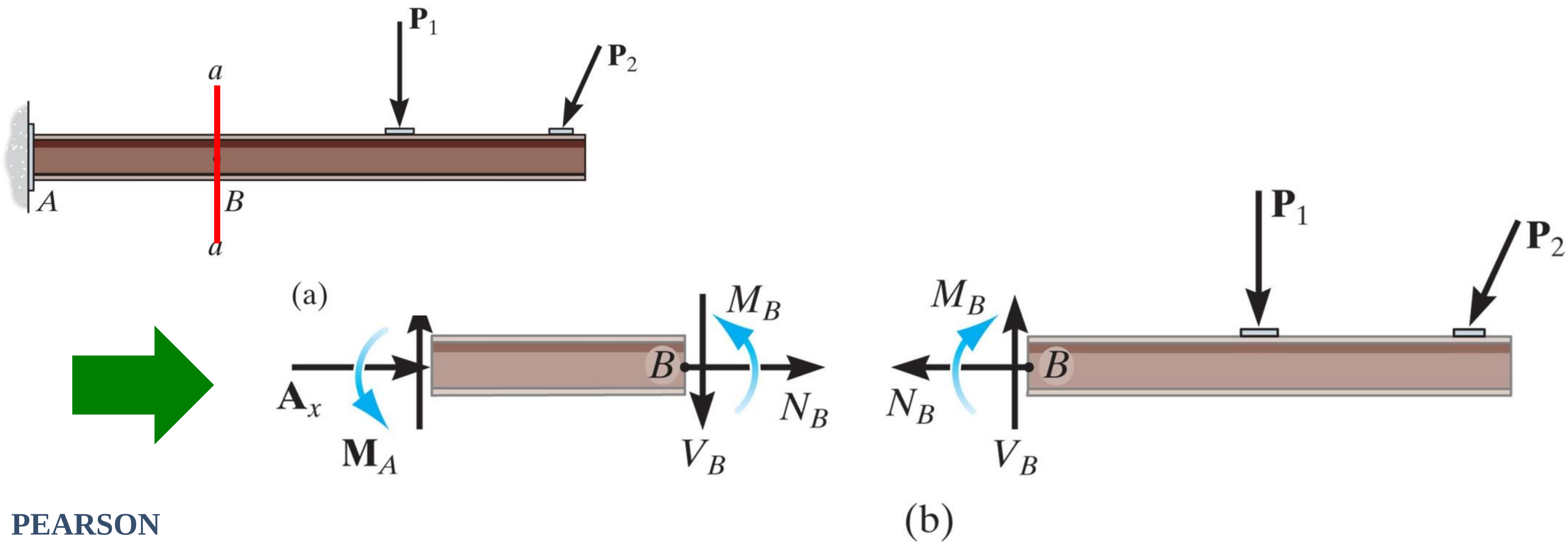
- ✓ Know the internal loading →make sure the material can resist the loading
- ✓ Internal loadings can be determined by using the *method of sections*



(a)

7.1 Internal Forces Developed in Structural Members

- ✓ Know the internal loading →make sure the material can resist the loading
- ✓ Internal loadings can be determined by using the *method of sections*

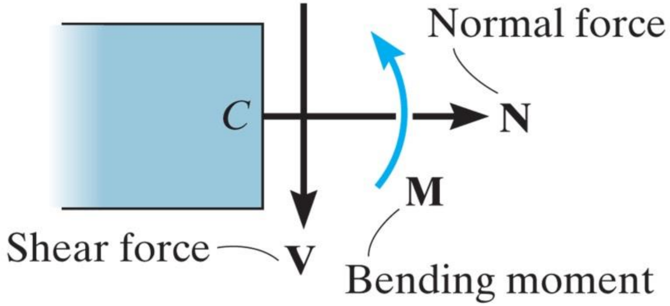


7.1 Internal
Struc

Figure: 07_PH001

In each case, the link on the backhoe is a two-force member. In the top photo it is subjected to both bending and an axial load at its center. By making the member straight, as in the bottom photo, then only an axial force acts within the member.

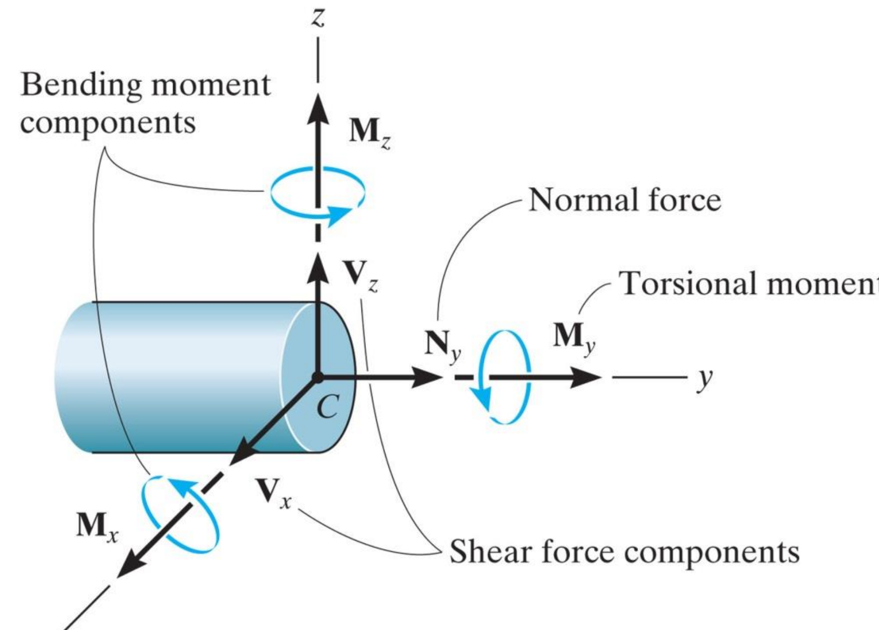
7.1 Internal Forces Developed in Structural Members



(a)

2D: 3 internal loading resultants

For 3D, a general internal force and couple moment resultant will act at the section



(b)

N_y : normal force

V_x , V_z : shear components

M_y : torisonal or twisting moment

M_x , M_z : bending moment components

Resultant loadings will act at the geometric center or centroid(C)

7.1 Internal Forces Developed in Structural Members

Sign Convention

- ✓ The normal force is said to be positive if it creates tension, a positive shear force will cause the beam segment on which it acts to rotate clockwise, and a positive bending moment will tend to bend the segment on which it acts in a concave upward manner.
- ✓ Loadings that are opposite to these are considered negative.

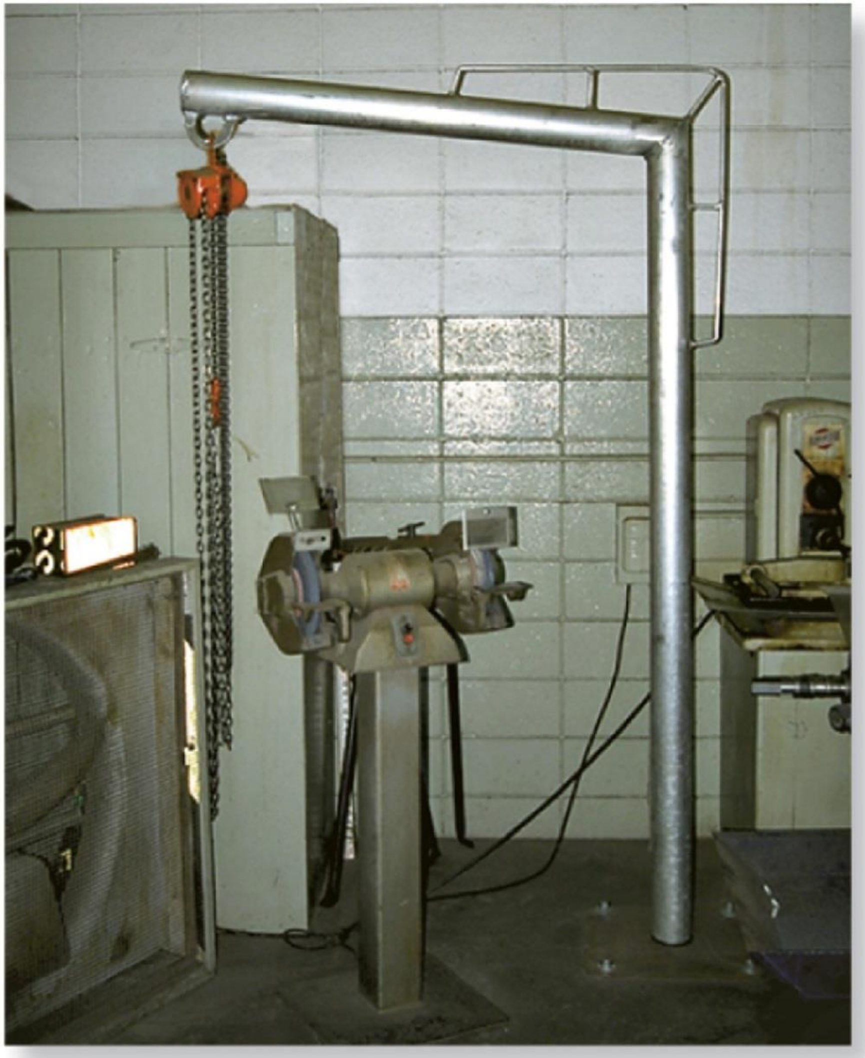
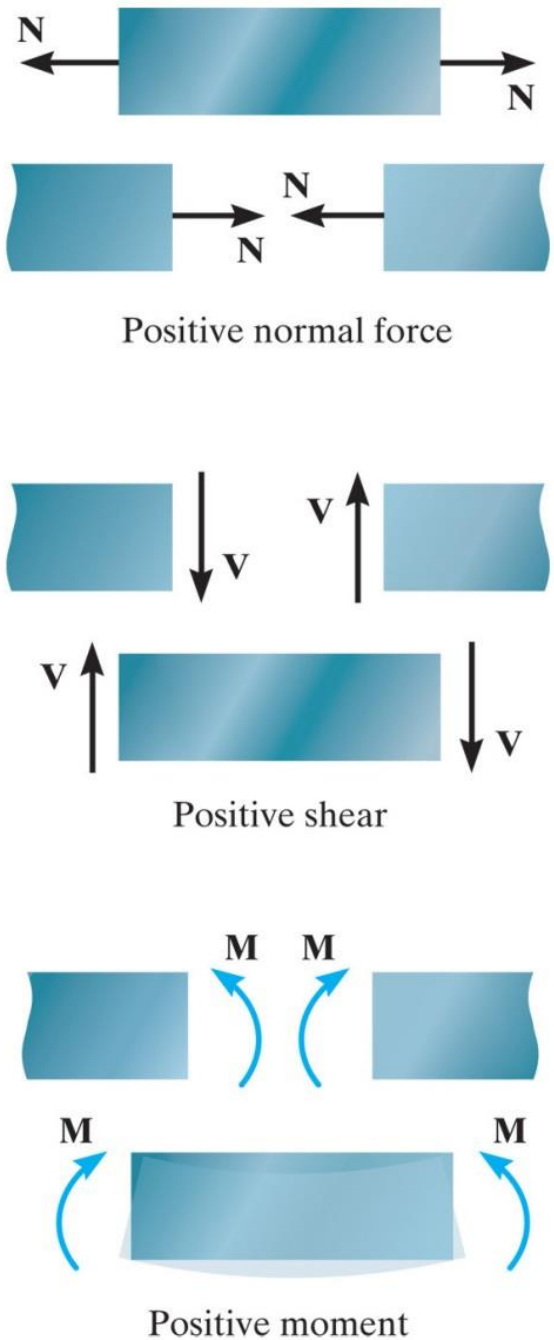


Figure: 07_PH002
The designer of this shop crane realized the need for additional reinforcement around the joint in order to prevent severe internal bending of the joint when a large load is suspended from the chain hoist.



Procedure for analysis

The method of sections can be used to determine the internal loadings on the cross section of a member using the following procedure.

Support Reactions.

- Before the member is sectioned, it may first be necessary to determine its support reactions.

Free-Body Diagram.

- It is important to *keep* all distributed loadings, couple moments, and forces acting on the member in their *exact locations*, *then* pass an imaginary section through the member, perpendicular to its axis at the point where the internal loadings are to be determined.

- After the section is made, draw a free-body diagram of the segment that has the least number of loads on it, and indicate the components of the internal force and couple moment resultants at the cross section acting in their positive directions in accordance with the established sign convention.

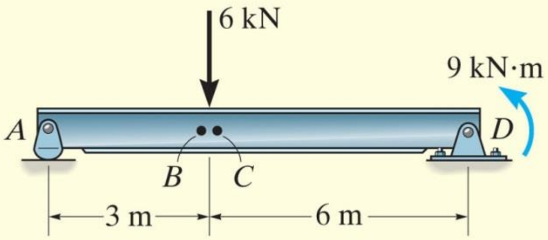
Procedure for analysis

Equations of Equilibrium.

- Moments should be summed at the section. This way the normal and shear forces at the section are eliminated, and we can obtain a direct solution for the moment.
- If the solution of the equilibrium equations yields a negative scalar, the sense of the quantity is opposite to that shown on the free-body diagram.

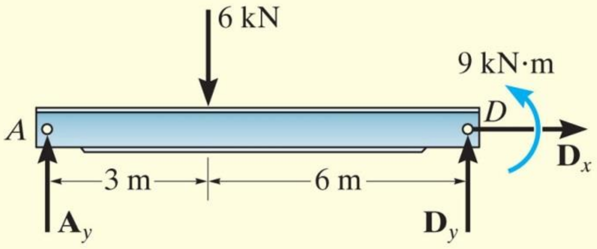
EXAMPLE 7.1

Determine the normal force, shear force, and bending moment acting just to the left, point B, and just to the right, point C, of the 6-kN force on the beam in Fig. 7–4*a*.



(a)

SOLUTION



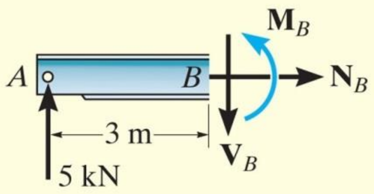
(b)

Support Reactions. The free-body diagram of the beam is shown in Fig. 7–4*b*. When determining the *external reactions*, realize that the 9-kN · m couple moment is a free vector and therefore it can be placed *anywhere* on the free-body diagram of the entire beam. Here we will only determine $\mathbf{A}_y$, since the left segments will be used for the analysis.

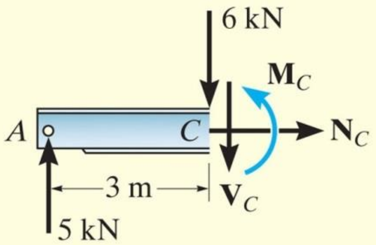
$$\circlearrowleft +\Sigma M_D = 0; \quad 9 \text{ kN} \cdot \text{m} + (6 \text{ kN})(6 \text{ m}) - A_y(9 \text{ m}) = 0$$

$$A_y = 5 \text{ kN}$$

EXAMPLE 7.1 CONTINUED



(c)



(d)

Fig. 7–4

Free-Body Diagrams. The free-body diagrams of the left segments AB and AC of the beam are shown in Figs. 7–4*c* and 7–4*d*. In this case the 9-kN · m couple moment is *not included* on these diagrams since it must be kept in its *original position* until *after* the section is made and the appropriate segment is isolated.

Equations of Equilibrium.

Segment AB

$$\xrightarrow{+} \Sigma F_x = 0; \quad N_B = 0 \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad 5\text{ kN} - V_B = 0 \quad V_B = 5\text{ kN} \quad \textit{Ans.}$$

$$\circlearrowleft + \Sigma M_B = 0; \quad -(5\text{ kN})(3\text{ m}) + M_B = 0 \quad M_B = 15\text{ kN}\cdot\text{m} \quad \textit{Ans.}$$

Segment AC

$$\xrightarrow{+} \Sigma F_x = 0; \quad N_C = 0 \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad 5\text{ kN} - 6\text{ kN} - V_C = 0 \quad V_C = -1\text{ kN} \quad \textit{Ans.}$$

$$\circlearrowleft + \Sigma M_C = 0; \quad -(5\text{ kN})(3\text{ m}) + M_C = 0 \quad M_C = 15\text{ kN}\cdot\text{m} \quad \textit{Ans.}$$

NOTE: The negative sign indicates that $\mathbf{V}_C$ acts in the opposite sense to that shown on the free-body diagram. Also, the moment arm for the 5-kN force in both cases is approximately 3 m since B and C are "almost" coincident.

EXAMPLE 7.2

Determine the normal force, shear force, and bending moment at C of the beam in Fig. 7–5a.

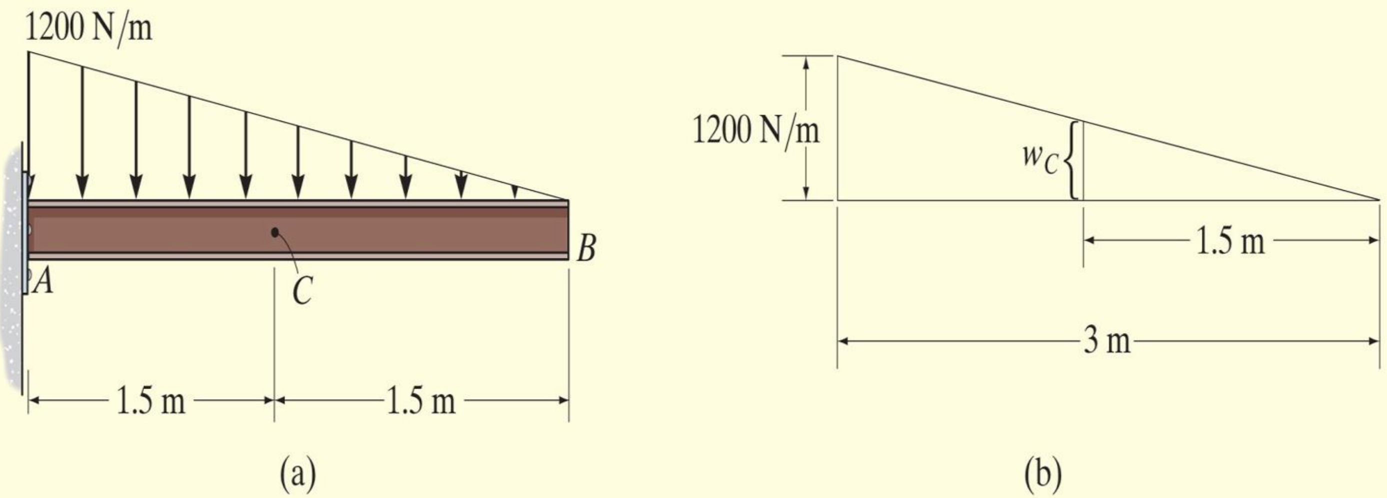


Fig. 7–5

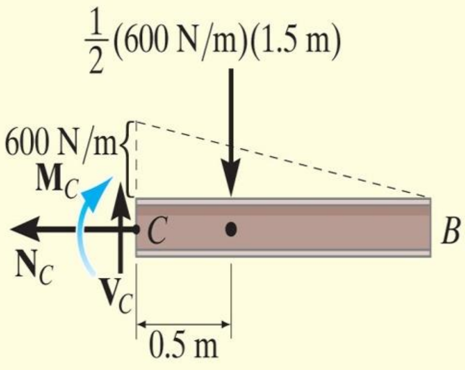
EXAMPLE 7.2 CONTINUED

SOLUTION

Free-Body Diagram. It is not necessary to find the support reactions at A since segment BC of the beam can be used to determine the internal loadings at C. The intensity of the triangular distributed load at C is determined using similar triangles from the geometry shown in Fig. 7–5b, i.e.,

$$w_C = (1200\ \text{N/m})\left(\frac{1.5\ \text{m}}{3\ \text{m}}\right) = 600\ \text{N/m}$$

The distributed load acting on segment BC can now be replaced by its resultant force, and its location is indicated on the free-body diagram, Fig. 7–5c.



(c)

EXAMPLE 7.2 CONTINUED

Equations of Equilibrium.

$$\xrightarrow{+} \Sigma F_x = 0; \qquad N_C = 0 \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \qquad V_C - \tfrac{1}{2}(600 \text{ N/m})(1.5 \text{ m}) = 0$$

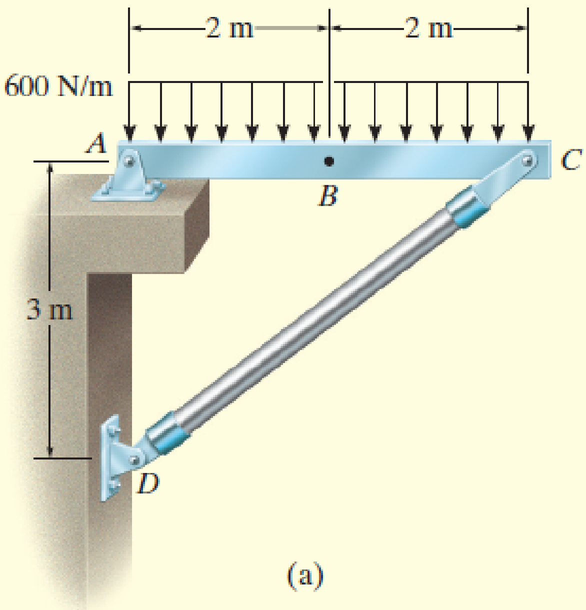
$$V_C = 450 \text{ N} \qquad \textit{Ans.}$$

$$\circlearrowleft + \Sigma M_C = 0; \quad -M_C - \tfrac{1}{2}(600 \text{ N/m})(1.5 \text{ m})(0.5 \text{ m}) = 0$$

$$M_C = -225 \text{ N} \qquad \textit{Ans.}$$

The negative sign indicates that $\mathbf{M}_C$ acts in the opposite sense to that shown on the free-body diagram.

EXAMPLE 7.3



(a)

Determine the normal force, shear force, and bending moment acting at point B of the two-member frame shown in Fig. 7–6*a*.

SOLUTION

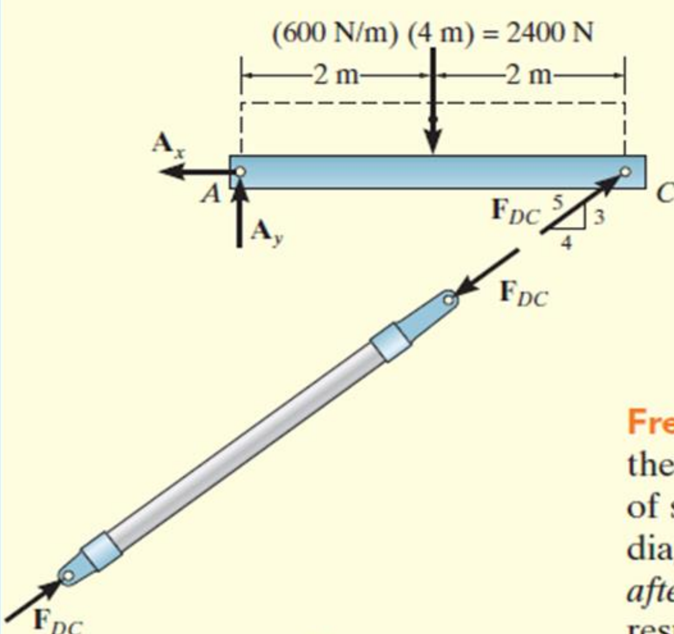
Support Reactions. A free-body diagram of each member is shown in Fig. 7–6*b*. Since CD is a two-force member, the equations of equilibrium need to be applied only to member AC.

$$\circlearrowleft +\Sigma M_A = 0; \quad -(2400 \text{ N})(2 \text{ m}) + \left(\tfrac{3}{5}\right) F_{DC}(4 \text{ m}) = 0 \quad F_{DC} = 2000 \text{ N}$$

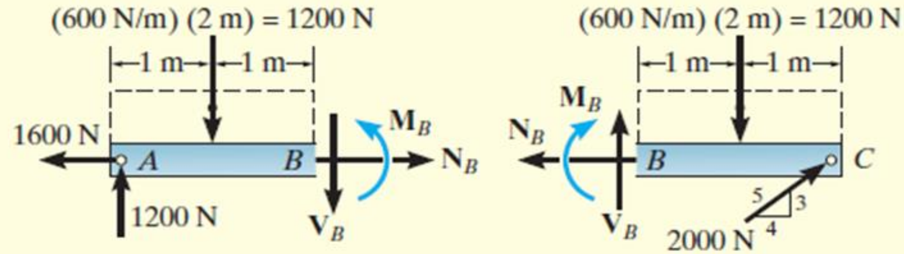
$$\xrightarrow{+} \Sigma F_x = 0; \quad -A_x + \left(\tfrac{4}{5}\right)(2000 \text{ N}) = 0 \quad A_x = 1600 \text{ N}$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - 2400 \text{ N} + \left(\tfrac{3}{5}\right)(2000 \text{ N}) = 0 \quad A_y = 1200 \text{ N}$$

EXAMPLE 7.3 CONTINUED



(b)



(c)

Fig. 7–6

Free-Body Diagrams. Passing an imaginary section perpendicular to the axis of member AC through point B yields the free-body diagrams of segments AB and BC shown in Fig. 7–6c. When constructing these diagrams it is important to keep the distributed loading where it is until *after the section is made*. Only then can it be replaced by a single resultant force.

Equations of Equilibrium. Applying the equations of equilibrium to segment AB, we have

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \quad N_B - 1600\text{ N} = 0 \qquad N_B = 1600\text{ N} = 1.60\text{ kN} \qquad \textit{Ans.}$$

$$+\uparrow\Sigma F_y = 0; 1200\text{ N} - 1200\text{ N} - V_B = 0 \qquad V_B = 0 \qquad \textit{Ans.}$$

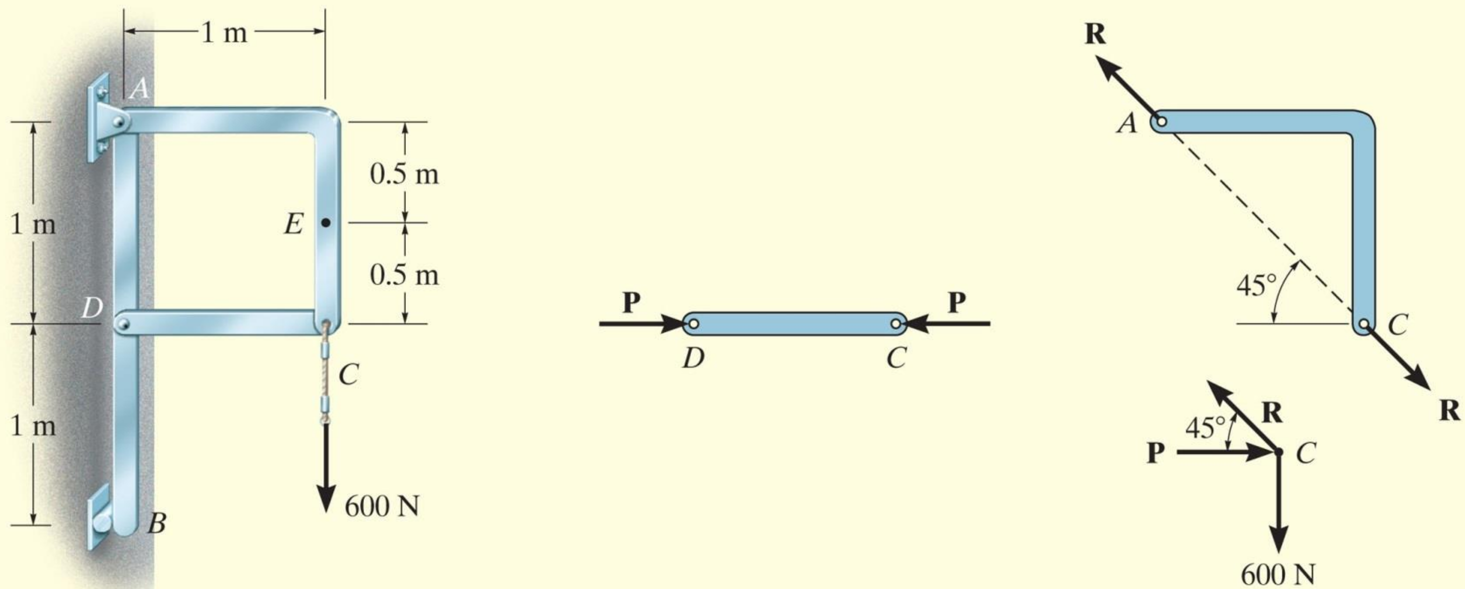
$$\circlearrowleft+\Sigma M_B = 0; \; M_B - 1200\text{ N (2 m)} + 1200\text{ N (1 m)} = 0$$

$$M_B = 1200\text{ N}\cdot\text{m} = 1.20\text{ kN}\cdot\text{m} \qquad \textit{Ans.}$$

NOTE: As an exercise, try to obtain these same results using segment BC.

EXAMPLE 7.4

Determine the normal force, shear force, and bending moment acting at point E of the frame loaded as shown in Fig. 7–7a.



(a) (b)

PEARSON

EXAMPLE 7.4 CONTINUED

SOLUTION

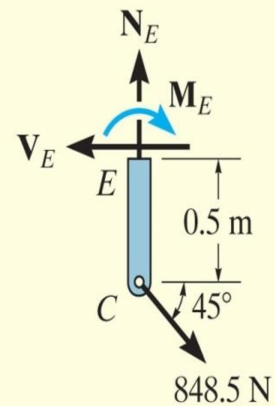
Support Reactions. By inspection, members AC and CD are two-force members, Fig. 7–7*b*. In order to determine the internal loadings at E, we must first determine the force $\mathbf{R}$ acting at the end of member AC. To obtain it, we will analyze the equilibrium of the pin at C.

Summing forces in the vertical direction on the pin, Fig. 7–7*b*, we have

$$+\uparrow \Sigma F_y = 0; \qquad R \sin 45^\circ - 600 \text{ N} = 0 \qquad R = 848.5 \text{ N}$$

EXAMPLE 7.4 CONTINUED

Free-Body Diagram. The free-body diagram of segment *CE* is shown in Fig. 7–7*c*.



(c)

Fig. 7–7

Equations of Equilibrium.

$$\xrightarrow{+} \Sigma F_x = 0; \qquad 848.5 \cos 45^\circ \text{ N} - V_E = 0 \qquad V_E = 600 \text{ N} \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \qquad -848.5 \sin 45^\circ \text{ N} + N_E = 0 \qquad N_E = 600 \text{ N} \qquad \textit{Ans.}$$

$$\circlearrowleft + \Sigma M_E = 0; \qquad 848.5 \cos 45^\circ \text{ N}(0.5 \text{ m}) - M_E = 0$$

$$M_E = 300 \text{ N} \cdot \text{m} \qquad \textit{Ans.}$$

NOTE: These results indicate a poor design. Member *AC* should be *straight* (from *A* to *C*) so that bending within the member is *eliminated*. If *AC* were straight then the internal force would only create tension in the member.

EXAMPLE 7.5

(© Russell C. Hibbeler)

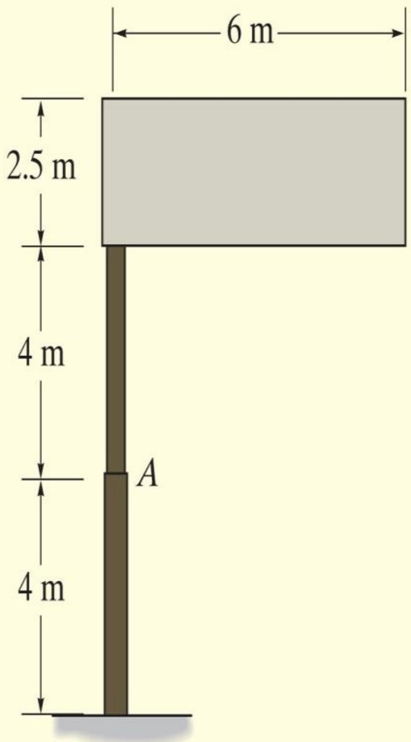
(a)

The uniform sign shown in Fig. 7–8*a* has a mass of 650 kg and is supported on the fixed column. Design codes indicate that the expected maximum uniform wind loading that will occur in the area where it is located is 900 Pa. Determine the internal loadings at *A*.

SOLUTION

The idealized model for the sign is shown in Fig. 7–8*b*. Here the necessary dimensions are indicated. We can consider the free-body diagram of a section above point *A* since it does not involve the support reactions.

EXAMPLE 7.5 CONTINUED



(b)

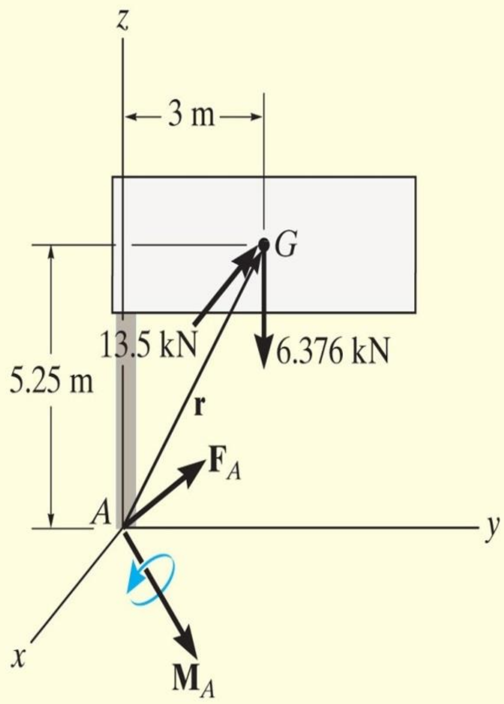
Free-Body Diagram. The sign has a weight of $W = 650(9.81)$ N $= 6.376$ kN, and the wind creates a resultant force of $F_w = 900\ \text{N/m}^2(6\ \text{m})(2.5\ \text{m}) = 13.5$ kN, which acts perpendicular to the face of the sign. These loadings are shown on the free-body diagram, Fig. 7–8*c*.

Equations of Equilibrium. Since the problem is three dimensional, a vector analysis will be used.

$$\Sigma \mathbf{F} = \mathbf{0}; \qquad \mathbf{F}_A - 13.5\mathbf{i} - 6.376\mathbf{k} = \mathbf{0}$$

$$\mathbf{F}_A = \{13.5\mathbf{i} + 6.38\mathbf{k}\}\ \text{kN} \qquad \textit{Ans.}$$

EXAMPLE 7.5 CONTINUED



(c)

Fig. 7–8

$\Sigma \mathbf{M}_A = \mathbf{0}; \qquad \mathbf{M}_A + \mathbf{r} \times (\mathbf{F}_w + \mathbf{W}) = \mathbf{0}$

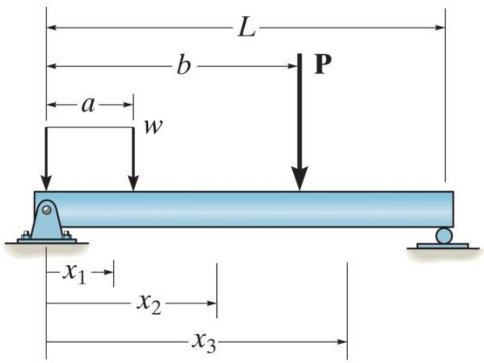
$$\mathbf{M}_A + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 5.25 \\ -13.5 & 0 & -6.376 \end{vmatrix} = \mathbf{0}$$

$$\mathbf{M}_A = \{19.1\mathbf{i} + 70.9\mathbf{j} - 40.5\mathbf{k}\}\ \text{kN}\cdot\text{m} \qquad \textit{Ans.}$$

NOTE: Here $\mathbf{F}_{A_z} = \{6.38\mathbf{k}\}$ kN represents the normal force, whereas $\mathbf{F}_{A_x} = \{13.5\mathbf{i}\}$ kN is the shear force. Also, the torsional moment is $\mathbf{M}_{A_z} = \{-40.5\mathbf{k}\}$ kN · m, and the bending moment is determined from its components $\mathbf{M}_{A_x} = \{19.1\mathbf{i}\}$ kN · m and $\mathbf{M}_{A_y} = \{70.9\mathbf{j}\}$ kN · m; i.e., $(M_b)_A = \sqrt{(M_A)_x^2 + (M_A)_y^2} = 73.4$ kN · m.

7.2 Shear and Moment Equations and Diagrams

- ✓ Beams are structural members designed to support loadings applied perpendicular to their axes
- ✓ A simply supported beam is pinned at one end and roller supported at the other
- ✓ A cantilevered beam is fixed at one end and free at the other



(a)

7.2 Shear and Moment Equations

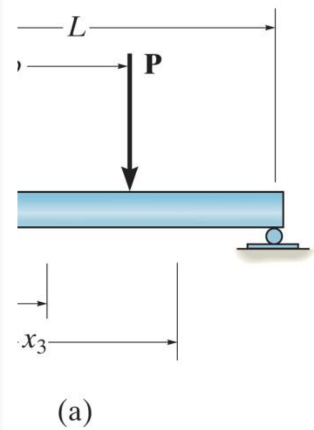
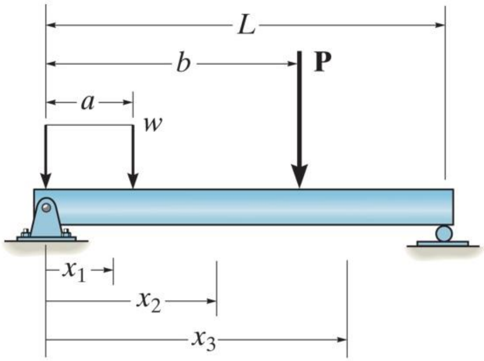


Figure: 07_PH003
To save on material and thereby produce an efficient design, these beams, also called girders, have been tapered, since the internal moment in the beam will be larger at the supports, or piers, than at the center of the span.

7.2 Shear and Moment Equations and Diagrams

- ✓ Internal shear and bending-moment functions will be discontinuous, or their slopes will be discontinuous, at points where a distributed load changes or where concentrated forces or couple moments are applied
- ✓ Functions must be determined for each segment of the beam located between any two discontinuities of loading

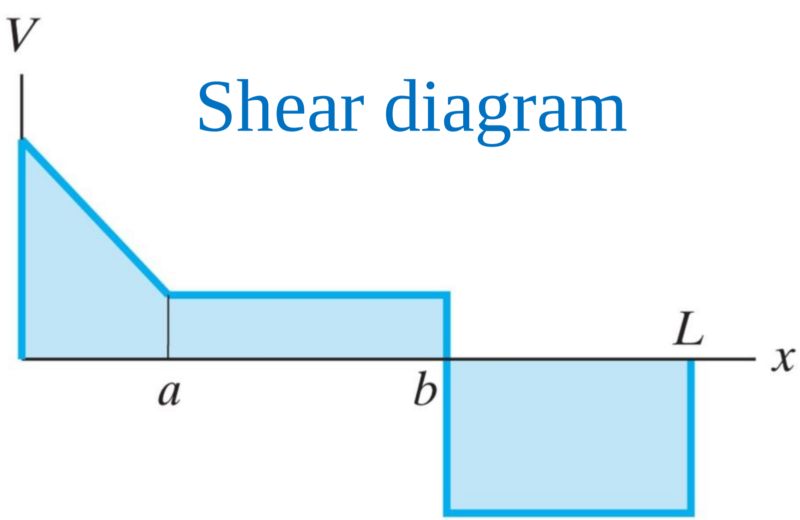
7.2 Shear and Moment Equations and Diagrams



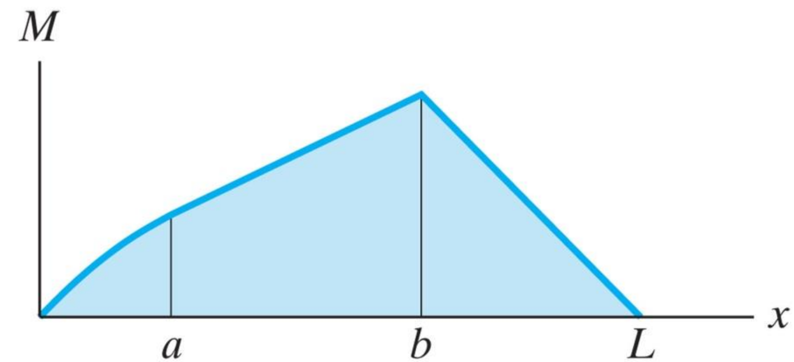
(a)

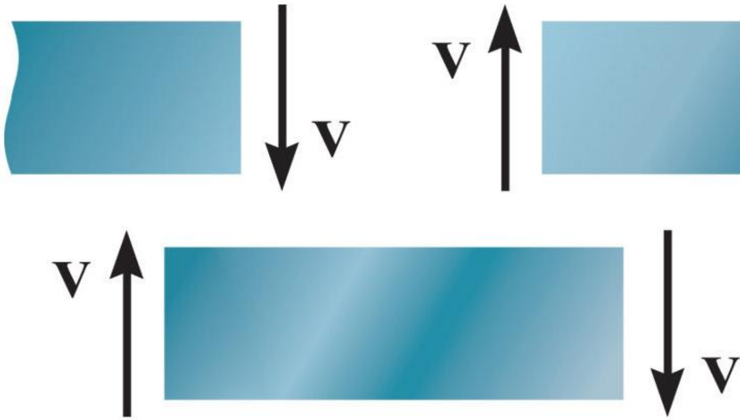
Internal normal force will not be considered

Shear diagram

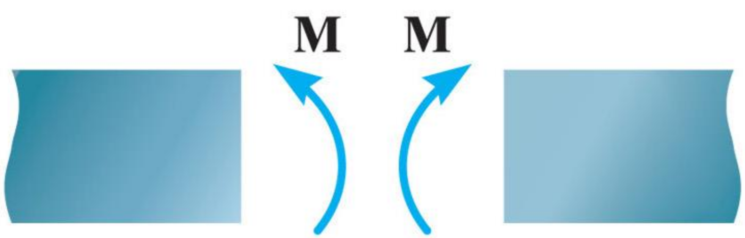


Bending-moment diagram





Positive shear



Positive moment

Beam sign convention

The shelving arms must be designed to resist the internal loading in the arms caused by the lumber.

Procedure for analysis

The shear and bending-moment diagrams for a beam can be constructed using the following procedure.

Support Reactions.

- Determine all the reactive forces and couple moments acting on the beam and resolve all the forces into components acting perpendicular and parallel to the beam's axis.

Procedure for analysis

Shear and Moment Functions.

- Specify separate coordinates x having an origin at the beam's left end and extending to regions of the beam *between* concentrated forces and/or couple moments, or where the distributed loading is continuous.

- Section the beam at each distance x and draw the free-body diagram of one of the segments. Be sure **V** and **M** are shown acting in their *positive sense*, in accordance with the sign convention given in Fig. 7–10.

- The shear V is obtained by summing forces perpendicular to the beam's axis, and the moment M is obtained by summing moments about the sectioned end of the segment.

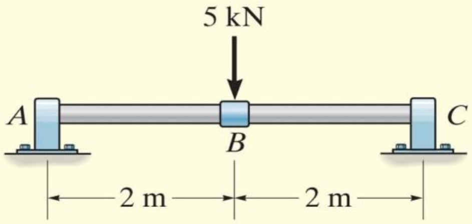
Procedure for analysis

Shear and Moment Diagrams.

- Plot the shear diagram (V versus x) and the moment diagram (M versus x). If computed values of the functions describing V and M are *positive*, the values are plotted above the x axis, whereas *negative* values are plotted below the x axis.

EXAMPLE 7.6

Draw the shear and moment diagrams for the shaft shown in Fig. 7–11*a*. The support at A is a thrust bearing and the support at C is a journal bearing.

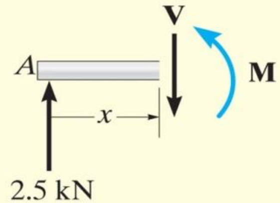


(a)

SOLUTION

Support Reactions. The support reactions are shown on the shaft's free-body diagram, Fig. 7–11*d*.

Shear and Moment Functions. The shaft is sectioned at an arbitrary distance x from point A, extending within the region AB, and the free-body diagram of the left segment is shown in Fig. 7–11*b*. The unknowns **V** and **M** are assumed to act in the *positive sense* on the right-hand face of the segment according to the established sign convention. Applying the equilibrium equations yields



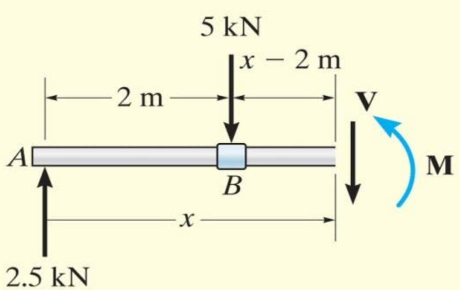
$0 \le x < 2$ m

(b)

$$+\uparrow \Sigma F_y = 0; \qquad V = 2.5 \text{ kN} \qquad (1)$$

$$\circlearrowleft + \Sigma M = 0; \qquad M = 2.5x \text{ kN}\cdot\text{m} \qquad (2)$$

A free-body diagram for a left segment of the shaft extending from A a distance x, within the region BC is shown in Fig. 7–11*c*. As always, **V** and **M** are shown acting in the positive sense. Hence,



$2 \text{ m} < x \le 4$ m

(c)

$$+\uparrow \Sigma F_y = 0; \qquad 2.5 \text{ kN} - 5 \text{ kN} - V = 0$$

$$V = -2.5 \text{ kN} \qquad (3)$$

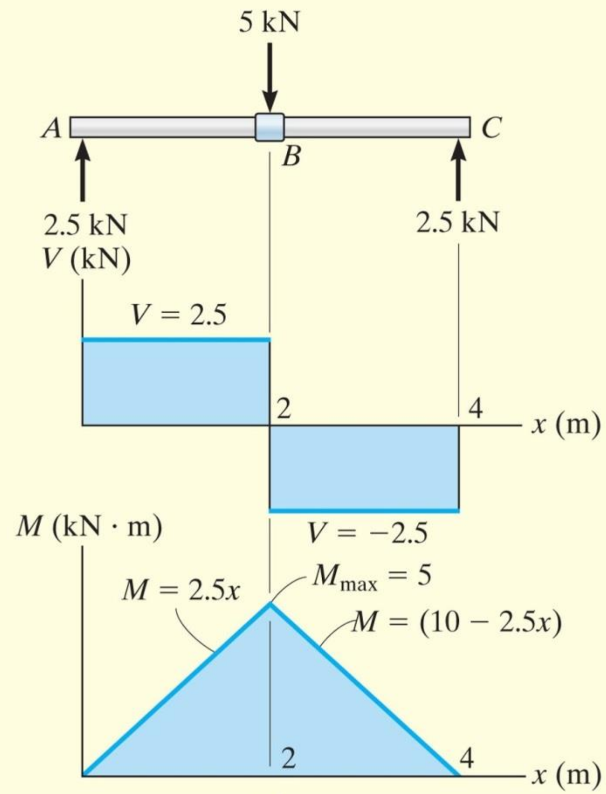
$$\circlearrowleft + \Sigma M = 0; \qquad M + 5 \text{ kN}(x - 2 \text{ m}) - 2.5 \text{ kN}(x) = 0$$

$$M = (10 - 2.5x) \text{ kN}\cdot\text{m} \qquad (4)$$

EXAMPLE 7.6 CONTINUED

Shear and Moment Diagrams. When Eqs. 1 through 4 are plotted within the regions in which they are valid, the shear and moment diagrams shown in Fig. 7–11*d* are obtained. The shear diagram indicates that the internal shear force is always 2.5 kN (positive) within segment AB. Just to the right of point B, the shear force changes sign and remains at a constant value of -2.5 kN for segment BC. The moment diagram starts at zero, increases linearly to point B at $x = 2$ m, where $M_{max} = 2.5\text{ kN}(2\text{ m}) = 5\text{ kN}\cdot\text{m}$, and thereafter decreases back to zero.

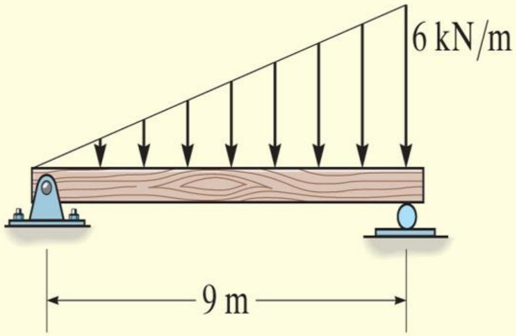
NOTE: It is seen in Fig. 7–11*d* that the graphs of the shear and moment diagrams "jump" or changes abruptly where the concentrated force acts, i.e., at points A, B, and C. For this reason, as stated earlier, it is necessary to express both the shear and moment functions separately for regions between concentrated loads. It should be realized, however, that all loading discontinuities are mathematical, arising from the *idealization of a concentrated force and couple moment*. Physically, loads are always applied over a finite area, and if the actual load variation could be accounted for, the shear and moment diagrams would then be continuous over the shaft's entire length.



(d)

Fig. 7–11

EXAMPLE 7.7



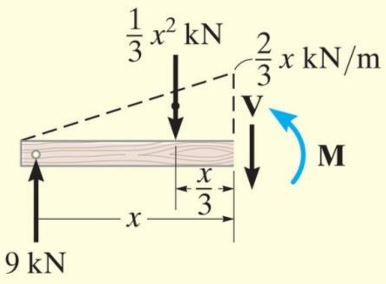
(a)

Draw the shear and moment diagrams for the beam shown in Fig. 7–12*a*.

SOLUTION

Support Reactions. The support reactions are shown on the beam's free-body diagram, Fig. 7–12*c*.

EXAMPLE 7.7 CONTINUED



(b)

Shear and Moment Functions. A free-body diagram for a left segment of the beam having a length x is shown in Fig. 7–12b. Due to proportional triangles, the distributed loading acting at the end of this segment has an intensity of $w/x = 6/9$ or $w = (2/3)x$. It is replaced by a resultant force *after* the segment is isolated as a free-body diagram. The *magnitude* of the resultant force is equal to $\frac{1}{2}(x)\left(\frac{2}{3}x\right) = \frac{1}{3}x^2$. This force *acts through the centroid* of the distributed loading area, a distance $\frac{1}{3}x$ from the right end. Applying the two equations of equilibrium yields

$$+\uparrow \Sigma F_y = 0; \qquad 9 - \frac{1}{3}x^2 - V = 0$$

$$V = \left(9 - \frac{x^2}{3}\right) \text{kN} \qquad (1)$$

$$\circlearrowleft + \Sigma M = 0; \qquad M + \frac{1}{3}x^2\left(\frac{x}{3}\right) - 9x = 0$$

$$M = \left(9x - \frac{x^3}{9}\right) \text{kN} \cdot \text{m} \qquad (2)$$

EXAMPLE 7.7 CONTINUED

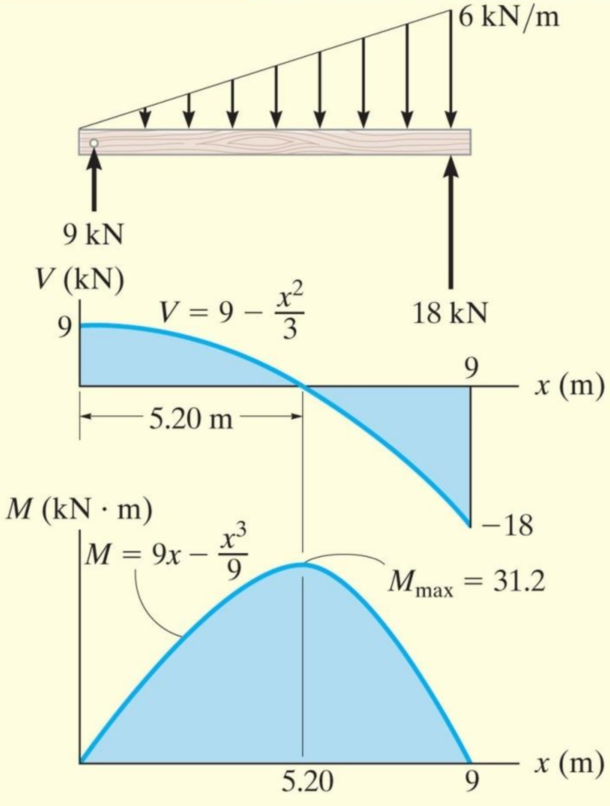


Fig. 7–12

Shear and Moment Diagrams. The shear and moment diagrams shown in Fig. 7–12*c* are obtained by plotting Eqs. 1 and 2.

The point of *zero shear* can be found using Eq. 1:

$$V = 9 - \frac{x^2}{3} = 0$$

$$x = 5.20 \text{ m}$$

NOTE: It will be shown in Sec. 7.3 that this value of x happens to represent the point on the beam where the *maximum moment* occurs. Using Eq. 2, we have

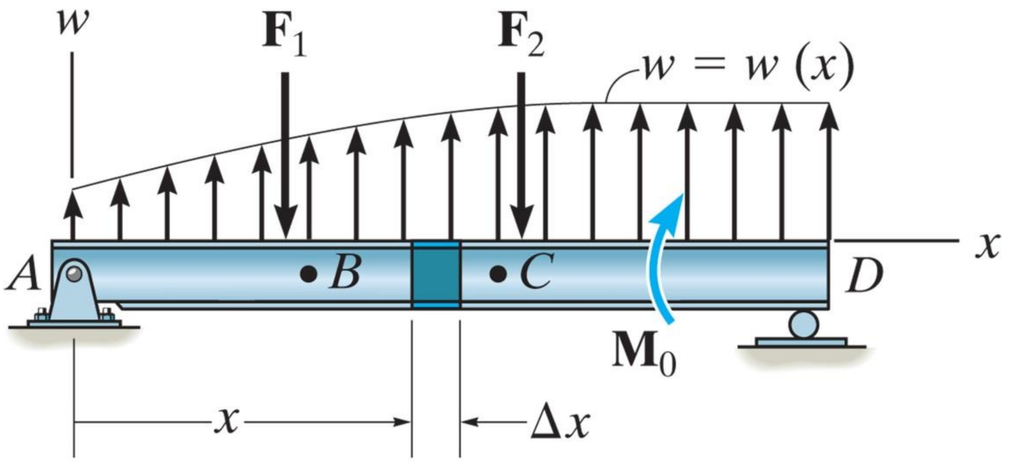
$$M_{max} = \left(9(5.20) - \frac{(5.20)^3}{9}\right) \text{kN} \cdot \text{m}$$

$$= 31.2 \text{ kN} \cdot \text{m}$$

7.3 Relations between Distributed Load, Shear and Moment

Distributed Load

- ✓ Consider beam AD subjected to an arbitrary load w = w(x) and a series of concentrated forces and moments
- ✓ Distributed load considered positive when the loading acts upwards



7.2 Relations between Distributed

ent

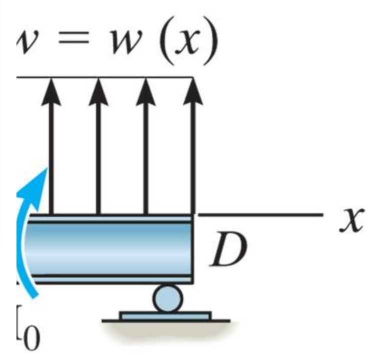
Figure: 07_PH005

In order to design the beam used to support these power lines, it is important to first draw the shear and moment diagrams for the beam.

ary

d

the

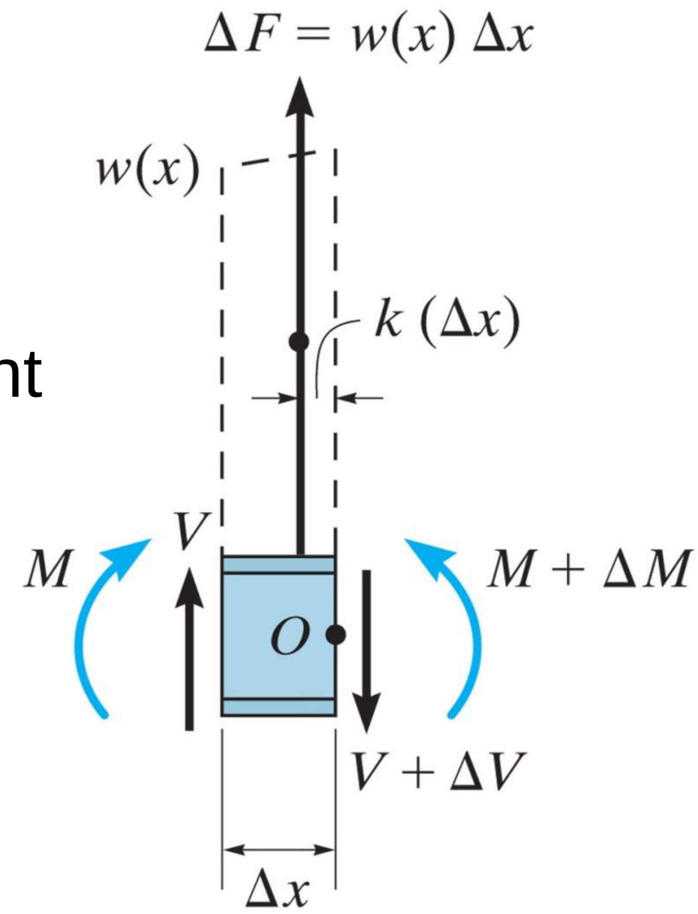


7.3 Relations between Distributed Load, Shear and Moment

- A FBD diagram for a small segment of the beam having a length Δx is chosen at point x along the beam which is not subjected to a concentrated force or couple moment

- Any results obtained will not apply at points of concentrated loading

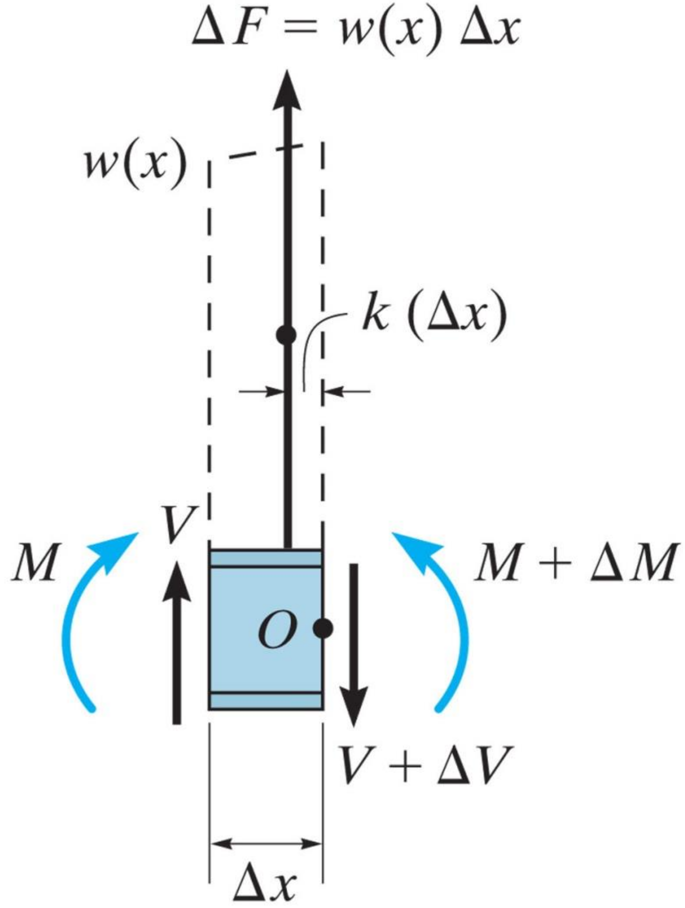
- The internal shear force and bending moments assumed positive sense



(b)

7.3 Relations between Distributed Load, Shear and Moment

- ✓ Distributed loading has been replaced by a resultant force ΔF = w(x) Δx that acts at a fractional distance k (Δx) from the right end, where 0 < k <1



(b)

7.3 Relations between Distributed Load, Shear and Moment

Relation between the distributed load and shear

$$\boxed{\begin{aligned} +\uparrow \Sigma F_y = 0; \qquad & V + w(x)\Delta x - (V + \Delta V) = 0 \\ & \Delta V = w(x)\Delta x \end{aligned}}$$

Dividing by Δx, and letting Δx→ 0

$$\frac{dV}{dx} = w(x)$$

$$\begin{matrix} \text{slope of} \\ \text{shear diagram} \end{matrix} = \begin{matrix} \text{distributed load} \\ \text{intensity} \end{matrix}$$

Rewrite: *dV=w(x)*

$$\Delta V = \int w(x)dx$$

$$\begin{matrix} \textbf{Change in} \\ \textbf{shear} \end{matrix} = \begin{matrix} \textbf{Area under} \\ \textbf{loading curve} \end{matrix}$$

7.3 Relations between Distributed Load, Shear and Moment

Relation Between the Shear and Moment

$$\circlearrowleft +\Sigma M_O = 0; \qquad (M + \Delta M) - [w(x)\Delta x]k\Delta x - V\Delta x - M = 0$$

$$\Delta M = V\Delta x + kw(x)\Delta x^2$$

Dividing by Δx, and letting Δx→ 0

$$\frac{dM}{dx} = V$$

$$\begin{array}{c}\text{slope of}\\ \text{moment diagram}\end{array} = \text{Shear}$$

Rewrite: *dM=∫*V dx

$$\Delta M = \int V\, dx$$

$$\begin{array}{c}\text{Change in}\\ \text{moment}\end{array} = \begin{array}{c}\text{Area under}\\ \text{shear diagram}\end{array}$$

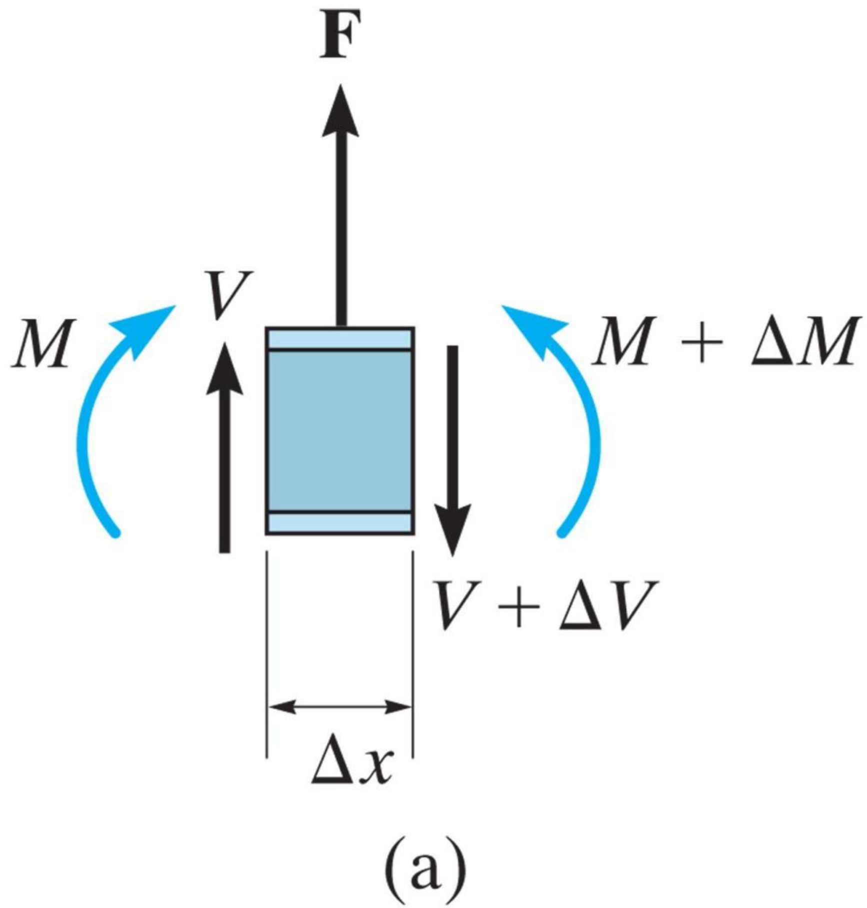
7.3 Relations between Distributed Load, Shear and Moment

Force

$$+\uparrow \Sigma F_y = 0; \qquad \boxed{\Delta V = F}$$

Since the change in shear is positive, the shear diagram will "jump" upward when F acts upward on the beam.

Likewise, the jump in shear (ΔV) is downward when F acts downward.



7.3 Relations between Distributed Load, Shear and Moment

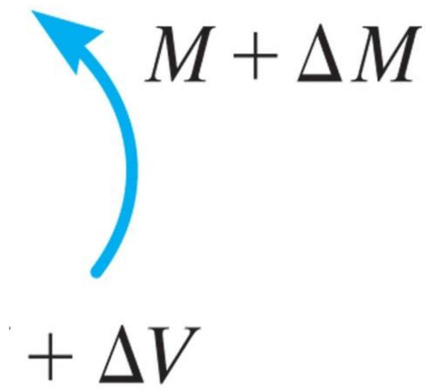
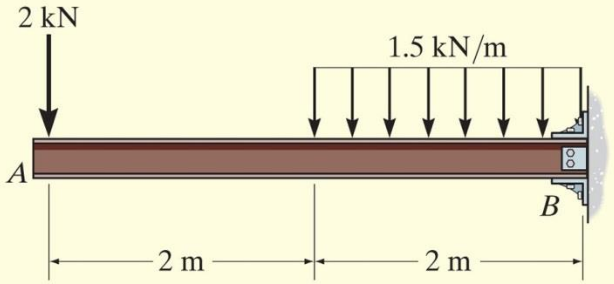


Figure: 07_PH006
This concrete beam is used to support the deck. Its size and the placement of steel reinforcement within it can be determined once the shear and moment diagrams have been established.

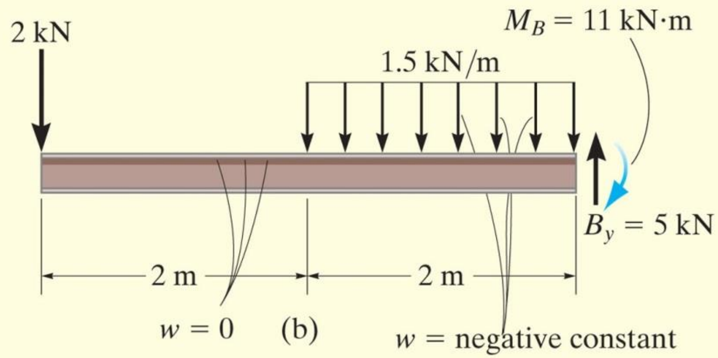
EXAMPLE 7.8

Draw the shear and moment diagrams for the cantilever beam in Fig. 7–15*a*.



(a)

Fig. 7–15



(b)

SOLUTION

The support reactions at the fixed support *B* are shown in Fig. 7–15*b*.

EXAMPLE 7.8 CONTINUED

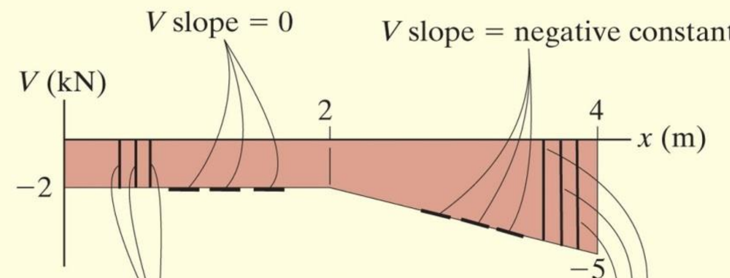
Shear Diagram. The shear at end A is -2 kN. This value is plotted at $x = 0$, Fig. 7–15*c*. Notice how the shear diagram is constructed by following the slopes defined by the loading w. The shear at $x = 4$ m is -5 kN, the reaction on the beam. This value can be verified by finding the area under the distributed loading; i.e.,

$$V|_{x=4\text{ m}} = V|_{x=2\text{ m}} + \Delta V = -2\text{ kN} - (1.5\text{ kN/m})(2\text{ m}) = -5\text{ kN}$$

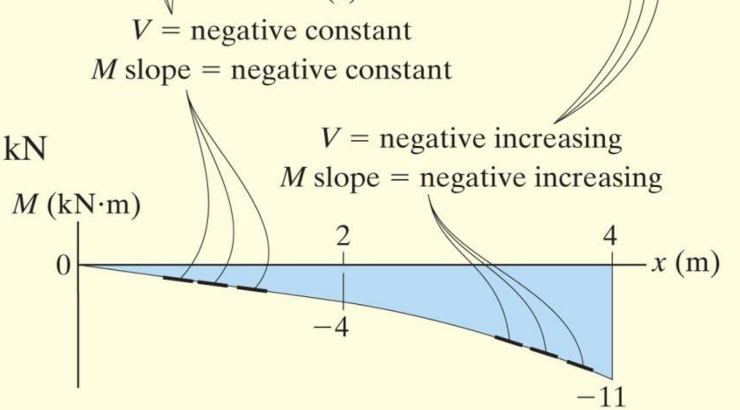
Moment Diagram. The moment of zero at $x = 0$ is plotted in Fig. 7–15*d*. Construction of the moment diagram is based on knowing that its slope is equal to the shear at each point. The change of moment from $x = 0$ to $x = 2$ m is determined from the area under the shear diagram. Hence, the moment at $x = 2$ m is

$$M|_{x=2\text{ m}} = M|_{x=0} + \Delta M = 0 + [-2\text{ kN}(2\text{ m})] = -4\text{ kN}\cdot\text{m}$$

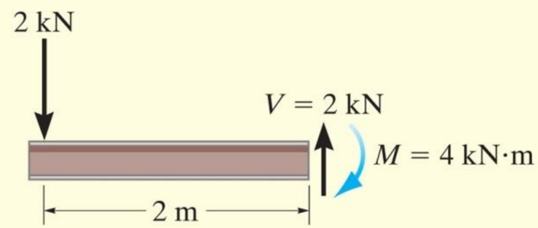
This same value can be determined from the method of sections, Fig. 7–15*e*.



(c)



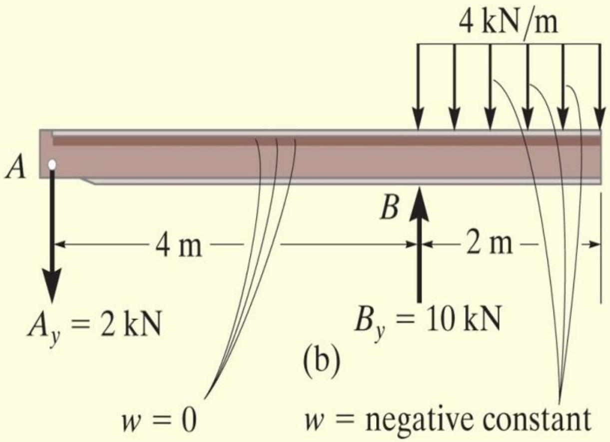
(d)



(e)

EXAMPLE 7.9

Draw the shear and moment diagrams for the overhang beam in Fig. 7–16*a*.



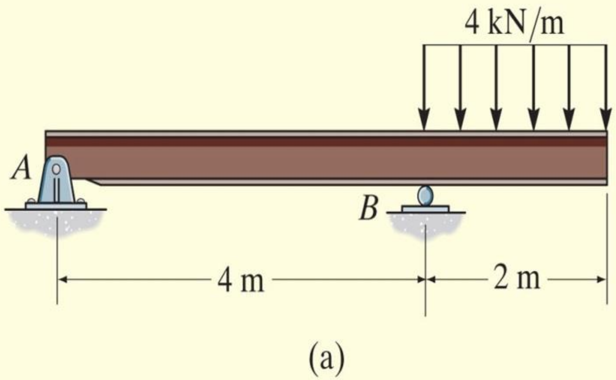


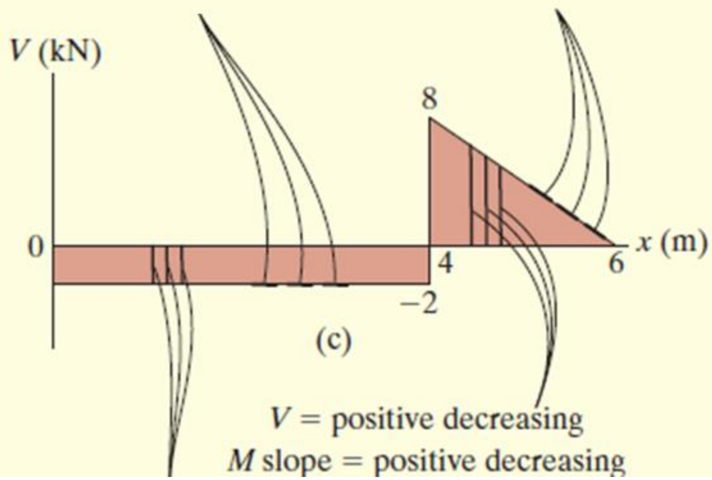
Fig. 7–16

SOLUTION

The support reactions are shown in Fig. 7–16*b*.

EXAMPLE 7.9 CONTINUED

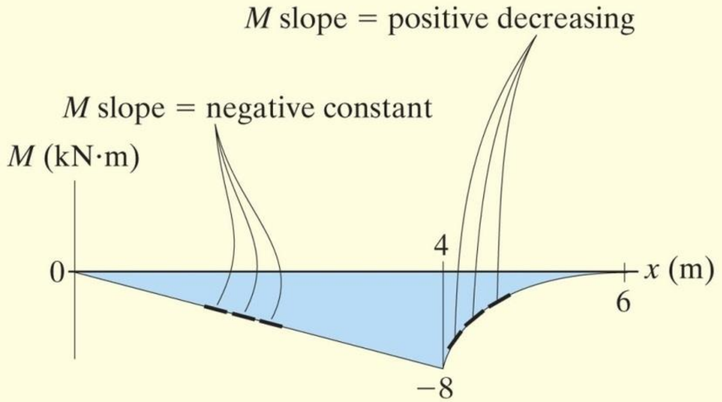
Shear Diagram. The shear of −2 kN at end A of the beam is plotted at $x = 0$, Fig. 7–16c. The slopes are determined from the loading and from this the shear diagram is constructed, as indicated in the figure. In particular, notice the positive jump of 10 kN at $x = 4$ m due to the force B_y, as indicated in the figure.



(c)

EXAMPLE 7.9 CONTINUED

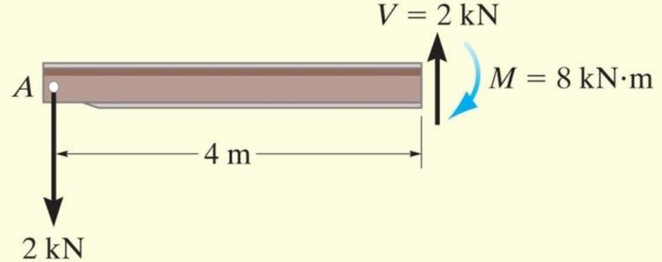
Moment Diagram. The moment of zero at $x = 0$ is plotted, Fig. 7–16*d*, then following the behavior of the slope found from the shear diagram, the moment diagram is constructed. The moment at $x = 4$ m is found from the area under the shear diagram.



(d)

$$M\big|_{x=4\text{ m}} = M\big|_{x=0} + \Delta M = 0 + [-2\text{ kN}(4\text{ m})] = -8\text{ kN}\cdot\text{m}$$

We can also obtain this value by using the method of sections, as shown in Fig. 7–16*e*.



(e)

EXAMPLE 7.10

The shaft in Fig. 7–17*a* is supported by a thrust bearing at *A* and a journal bearing at *B*. Draw the shear and moment diagrams.

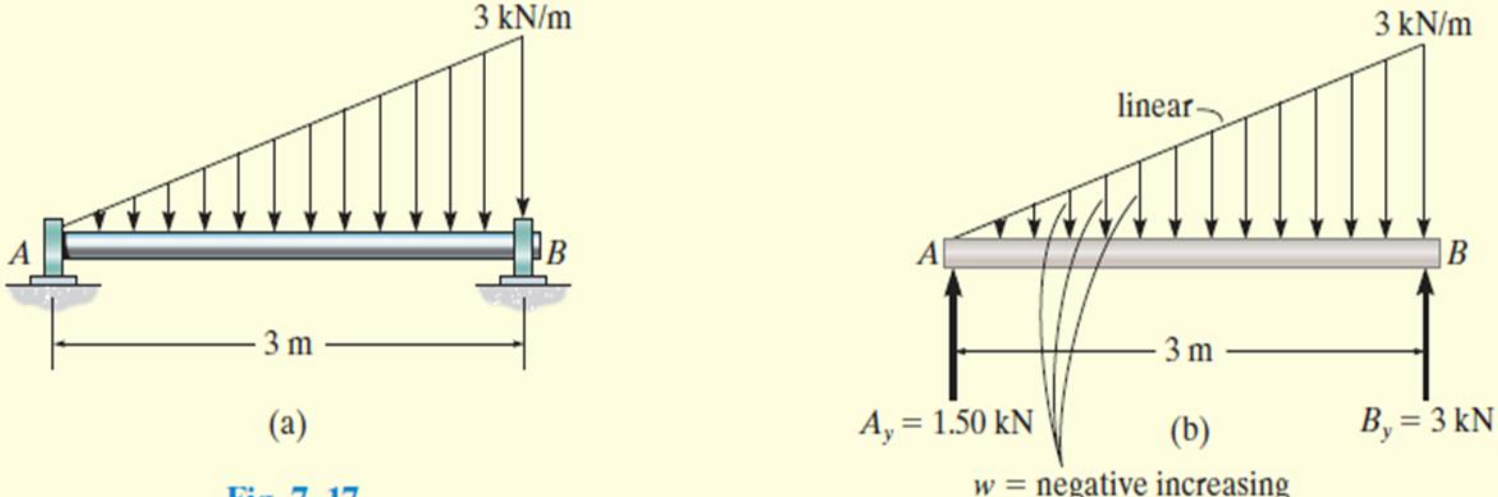


Fig. 7–17

EXAMPLE 7.10 CONTINUED

SOLUTION

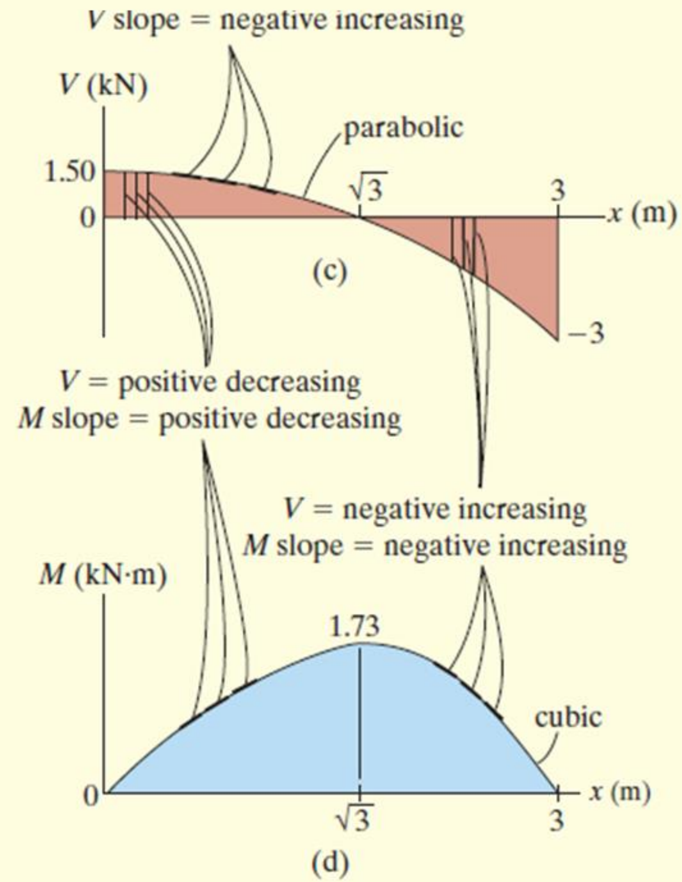
The support reactions are shown in Fig. 7–17*b*.

Shear Diagram. As shown in Fig. 7–17*c*, the shear at $x = 0$ is $+1.50$ kN. Following the slope defined by the loading, the shear diagram is constructed, where at *B* its value is -3 kN. Since the shear changes sign, the point where $V = 0$ must be located. To do this we will use the method of sections. The free-body diagram of the left segment of the shaft, sectioned at an arbitrary position x within the region $0 \le x < 3$ m, is shown in Fig. 7–17*e*. Notice that the intensity of the distributed load at x is $w = x$, which has been found by proportional triangles, i.e., $3/3 = w/x$.

Thus, for $V = 0$,

$$+\uparrow \Sigma F_y = 0; \qquad 1.50 \text{ kN} - \tfrac{1}{2}(x)(x) = 0$$

$$x = \sqrt{3} \text{ m}$$

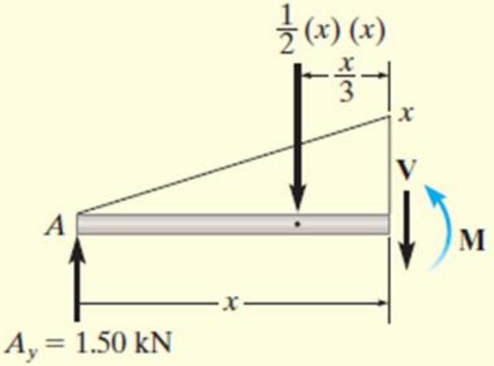


EXAMPLE 7.10 CONTINUED

Moment Diagram. The moment diagram starts at 0 since there is no moment at A, then it is constructed based on the slope as determined from the shear diagram. The maximum moment occurs at $x = \sqrt{3}$ m, where the shear is equal to zero, since $dM/dx = V = 0$, Fig. 7–17e,

$$\zeta + \Sigma M = 0; \quad M_{max} + \left[\tfrac{1}{2}(\sqrt{3})(\sqrt{3})\right]\left(\frac{\sqrt{3}}{3}\right) - 1.50(\sqrt{3}) = 0$$

$$M_{max} = 1.73 \text{ kN} \cdot \text{m}$$



(e)

Finally, notice how integration, first of the loading w which is linear, produces a shear diagram which is parabolic, and then a moment diagram which is cubic.

7.4 Cables

- ✓ Cables and chains used to support and transmit loads from one member to another
- ✓ In the force analysis, the weight of cables is neglected
- ✓ Assume cable is perfectly flexible and inextensible
  - Flexible: the cable offers no resistance to bending
  - Inextensible: Length remains constant before and after loading

ner

s is

nding

e and

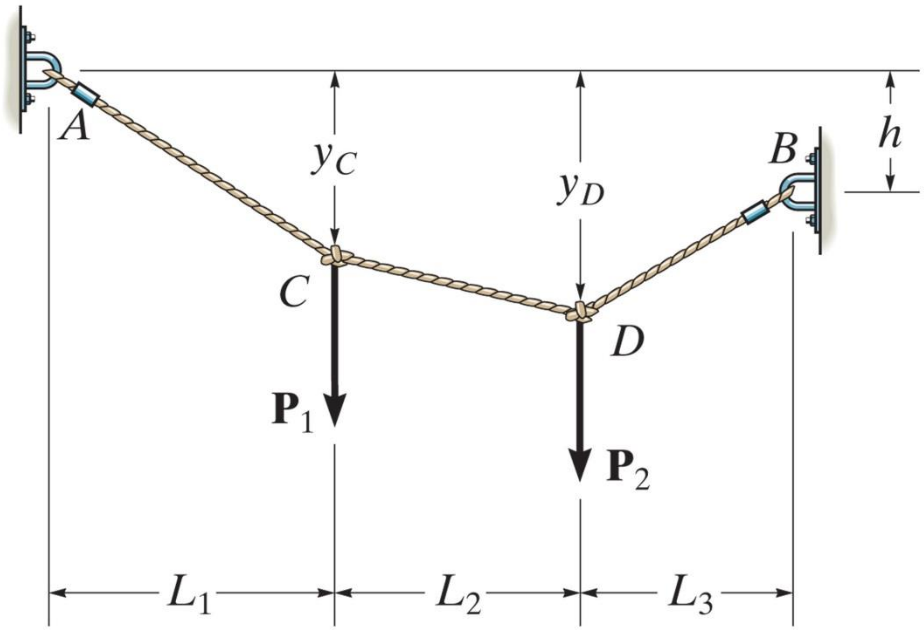
Figure: 07_PH007

Each of the cable segments remains approximately straight as they support the weight of these traffic lights.

7.4 Cables

Cable Subjected to Concentrated Loads

For a cable of negligible weight supporting several concentrated loads, the cable takes the form of several straight line segments, each subjected to constant tensile force



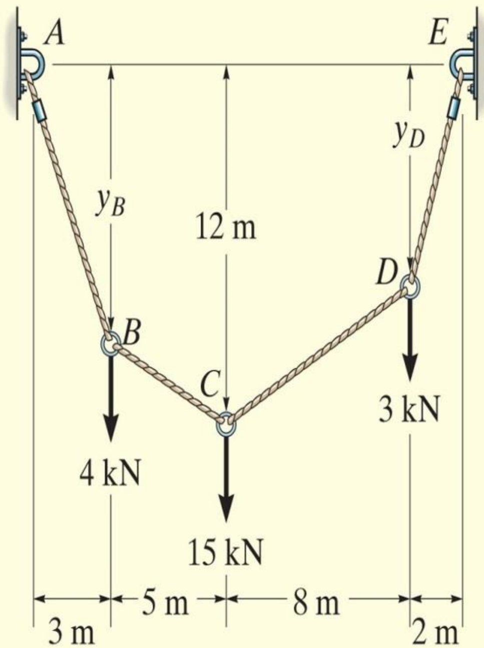
7.4 Cables

Cable Subjected to Concentrated Loads

- Known: h, L_1, L_2, L_3 and loads $\mathbf{P}_1$ and $\mathbf{P}_2$
- Form 2 equations of force equilibrium at each point A, B, C and D
- If the total length L is specified, use Pythagorean Theorem to relate the three segmental lengths
- If not, specify one of the sags, y_C or y_D, and from the answer determine the other sag and hence, the total length L

EXAMPLE 7.11

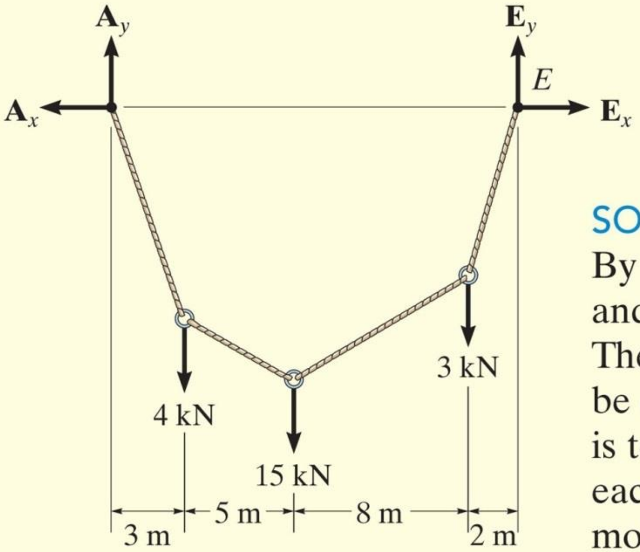
Determine the tension in each segment of the cable shown in Fig. 7–19a.



(a)

PEARSON

EXAMPLE 7.11 CONTINUED



(b)

SOLUTION

By inspection, there are four unknown external reactions (A_x, A_y, E_x, and E_y) and four unknown cable tensions, one in each cable segment. These eight unknowns along with the two unknown sags y_B and y_D can be determined from *ten* available equilibrium equations. One method is to apply the force equations of equilibrium ($\Sigma F_x = 0$, $\Sigma F_y = 0$) to each of the five points A through E. Here, however, we will take a more direct approach.

Consider the free-body diagram for the entire cable, Fig. 7–19*b*. Thus,

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \qquad -A_x + E_x = 0$$

$$\circlearrowleft +\Sigma M_E = 0;$$

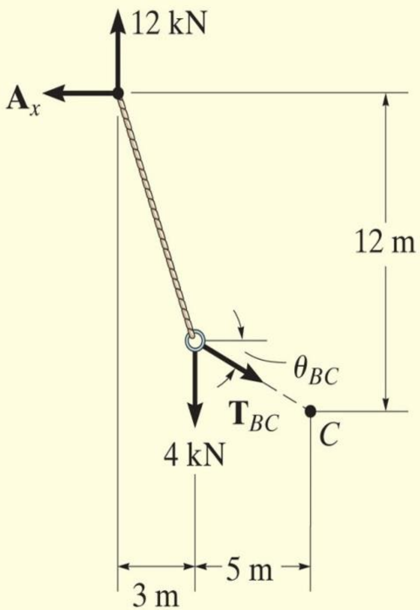
$$-A_y(18 \text{ m}) + 4 \text{ kN}\,(15 \text{ m}) + 15 \text{ kN}\,(10 \text{ m}) + 3 \text{ kN}\,(2 \text{ m}) = 0$$

$$A_y = 12 \text{ kN}$$

$$+\uparrow\Sigma F_y = 0; \qquad 12 \text{ kN} - 4 \text{ kN} - 15 \text{ kN} - 3 \text{ kN} + E_y = 0$$

$$E_y = 10 \text{ kN}$$

EXAMPLE 7.11 CONTINUED



(c)

Fig. 7–19

Since the sag $y_C = 12$ m is known, we will now consider the leftmost section, which cuts cable BC, Fig. 7–19*c*.

$$\circlearrowleft + \Sigma M_C = 0;\ A_x(12\text{ m}) - 12\text{ kN }(8\text{ m}) + 4\text{ kN }(5\text{ m}) = 0$$

$$A_x = E_x = 6.33\text{ kN}$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad T_{BC}\cos\theta_{BC} - 6.33\text{ kN} = 0$$

$$+\uparrow \Sigma F_y = 0; \quad 12\text{ kN} - 4\text{ kN} - T_{BC}\sin\theta_{BC} = 0$$

Thus,

$$\theta_{BC} = 51.6^\circ$$

$$T_{BC} = 10.2\text{ kN} \qquad \textit{Ans.}$$

EXAMPLE 7.11 CONTINUED

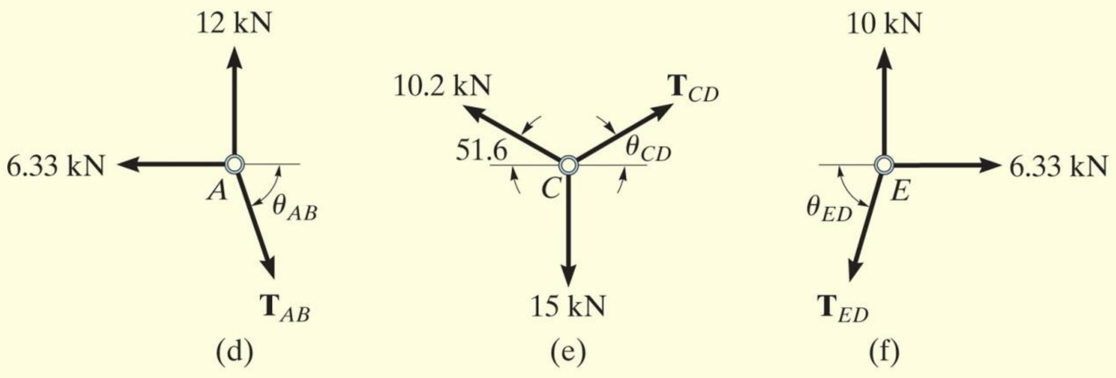


Fig. 7–19 (cont.)

Proceeding now to analyze the equilibrium of points A, C, and E in sequence, we have

Point A. (Fig. 7–19d).

$$\xrightarrow{+} \Sigma F_x = 0; \qquad T_{AB} \cos \theta_{AB} - 6.33 \text{ kN} = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad -T_{AB} \sin \theta_{AB} + 12 \text{ kN} = 0$$

$$\theta_{AB} = 62.2°$$

$$T_{AB} = 13.6 \text{ kN} \qquad \textit{Ans.}$$

EXAMPLE 7.11 CONTINUED

Point C. (Fig. 7–19*e*).

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \qquad T_{CD}\cos\theta_{CD} - 10.2\cos 51.6^\circ\text{ kN} = 0$$

$$+\uparrow\Sigma F_y = 0; \qquad T_{CD}\sin\theta_{CD} + 10.2\sin 51.6^\circ\text{ kN} - 15\text{ kN} = 0$$

$$\theta_{CD} = 47.9^\circ$$

$$T_{CD} = 9.44\text{ kN} \qquad \textit{Ans.}$$

Point E. (Fig. 7–19*f*).

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \qquad 6.33\text{ kN} - T_{ED}\cos\theta_{ED} = 0$$

$$+\uparrow\Sigma F_y = 0; \qquad 10\text{ kN} - T_{ED}\sin\theta_{ED} = 0$$

$$\theta_{ED} = 57.7^\circ$$

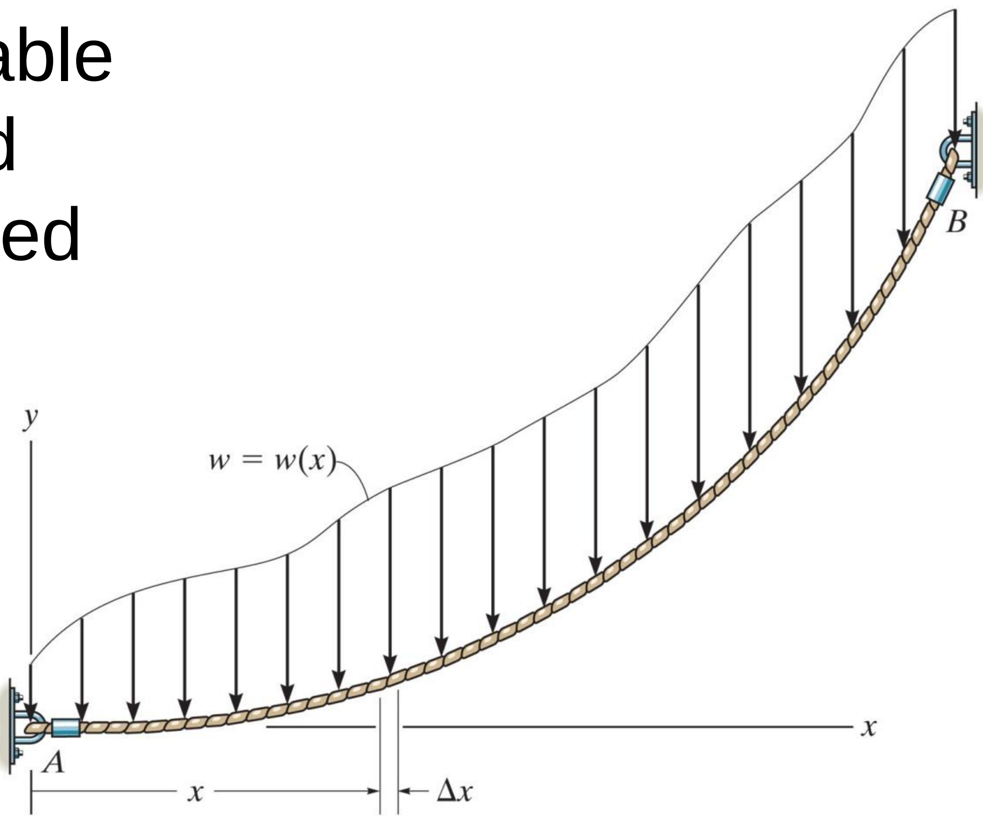
$$T_{ED} = 11.8\text{ kN} \qquad \textit{Ans.}$$

NOTE: By comparison, the maximum cable tension is in segment AB since this segment has the greatest slope (θ) and it is required that for any cable segment the horizontal component $T\cos\theta = A_x = E_x$ (a constant). Also, since the slope angles that the cable segments make with the horizontal have now been determined, it is possible to determine the sags y_B and y_D, Fig. 7–19*a*, using trigonometry.

7.4 Cables

Cable Subjected to a Distributed Load

✓Consider weightless cable subjected to a distributed loading w = w(x) measured in the x direction



(a)

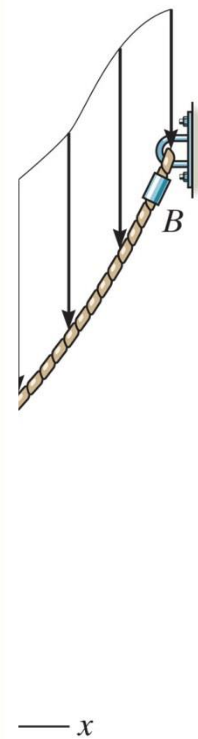


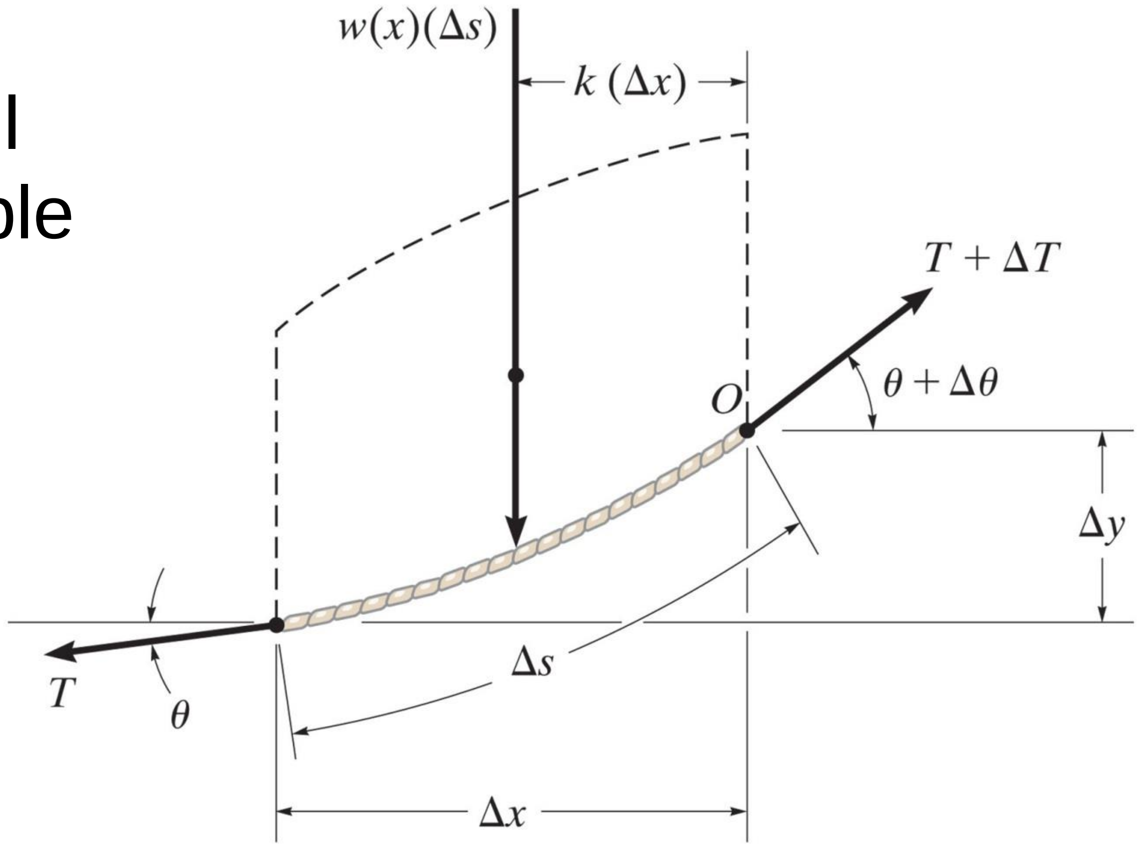
Figure: 07_PH008

The cable and suspenders are used to support the uniform load of a gas pipe which crosses the river.

7.4 Cables

Cable Subjected to a Distributed Load

The FBD of a small segment of the cable having a length Δs



(b)

7.4 Cables

Cable Subjected to a Distributed Load

- ✓ Since the tensile force in the cable changes in both magnitude and direction along the cable's length, this change is denoted on the FBD by ΔT

- ✓ Distributed load is represented by its resultant force $w(x)(\Delta x)$ which acts a fractional distance $k(\Delta x)$ from point O where o < k < 1

Cable Subjected to a Distributed Load

$$+\rightarrow \sum F_x = 0;$$

$$-T\cos\theta + (T+\Delta T)\cos(\theta+\Delta\theta) = 0$$

$$+\uparrow \sum F_y = 0;$$

$$-T\sin\theta - w(x)(\Delta x) + (T+\Delta T)\sin(\theta+\Delta\theta) = 0$$

$$\sum M_O = 0;$$

$$w(x)(\Delta x)k(\Delta x) - T\cos\theta\Delta y + T\sin\theta\Delta x = 0$$

7.4 Cables

Dividing each of these equations by Δx and taking the limit as Δx → 0, and therefore Δy →0, ΔΘ → 0, and ΔT→ 0, we obtain

$$\frac{d(T\cos\theta)}{dx} = 0 \qquad (7\text{-}7)$$

$$\frac{d(T\sin\theta)}{dx} - w(x) = 0 \qquad (7\text{-}8)$$

$$\frac{dy}{dx} = \tan\theta \qquad (7\text{-}9)$$

Integrating Eq.7-7

$$T\cos\theta = \text{constant} = F_H \qquad \textbf{(7-10)}$$

Integrating Eq.7-8

$$T\sin\theta = \int w(x)dx \qquad \textbf{(7-11)}$$

7.4 Cables

Dividing Eq. 7-11 by Eq. 7-10 eliminates T. Then, using Eq. 7-9, we can obtain the slope of the cable.

$$\tan\theta = \frac{dy}{dx} = \frac{1}{F_H}\int w(x)dx$$

Performing a second integration yields

$$y = \frac{1}{F_H}\int\left(\int w(x)dx\right)dx$$

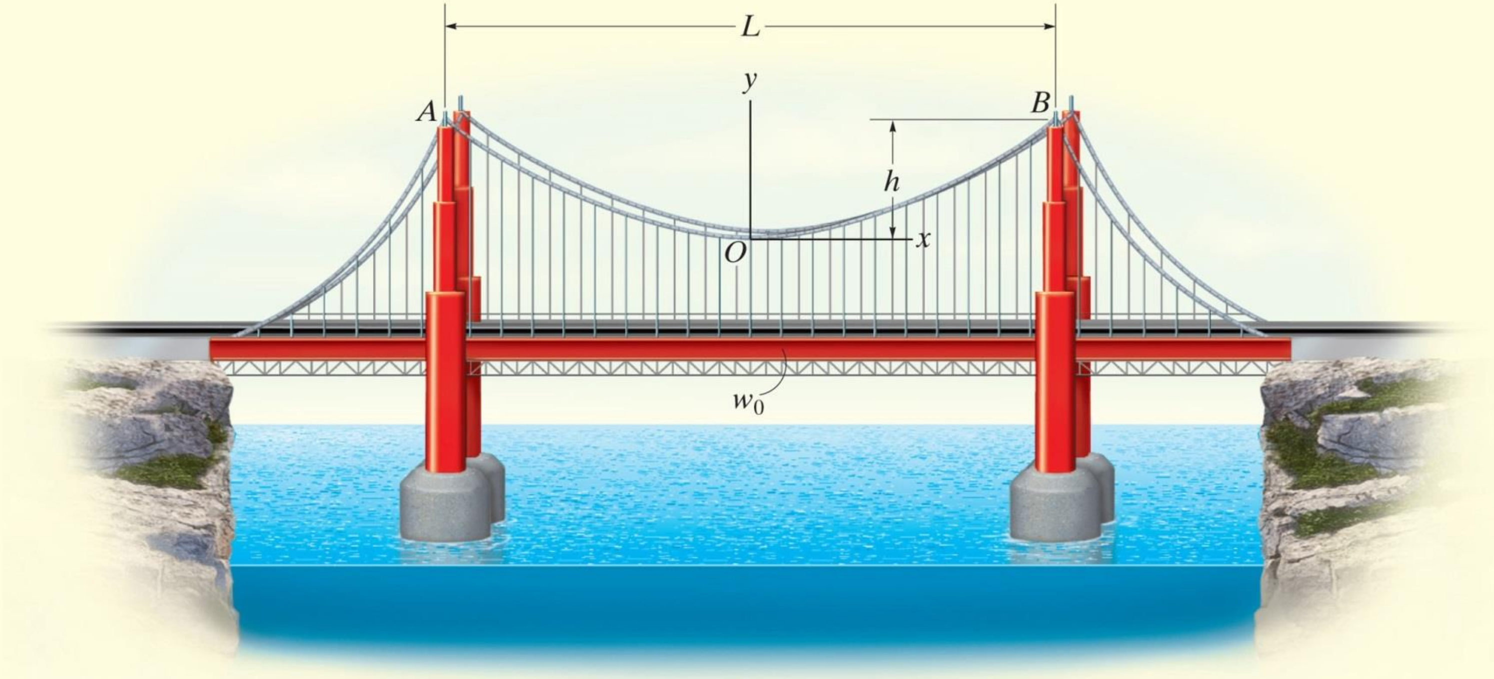
Dividing E
using Eq.

Performin

Figure: 07_PH009
The cables of the suspension bridge exert very large forces on the tower and the foundation block which have to be accounted for in their design.

EXAMPLE 7.12

The cable of a suspension bridge supports half of the uniform road surface between the two towers at A and B, Fig. 7–21a. If this distributed loading is w_0, determine the maximum force developed in the cable and the cable's required length. The span length L and sag h are known.



(a)

Fig. 7–21

EXAMPLE 7.12 CONTINUED

SOLUTION

We can determine the unknowns in the problem by first finding the equation of the curve that defines the shape of the cable using Eq. 7–12. For reasons of symmetry, the origin of coordinates has been placed at the cable's center. Noting that $w(x) = w_0$, we have

$$y = \frac{1}{F_H}\int\left(\int w_0\,dx\right)dx$$

Performing the two integrations gives

$$y = \frac{1}{F_H}\left(\frac{w_0 x^2}{2} + C_1 x + C_2\right) \qquad (1)$$

The constants of integration may be determined using the boundary conditions $y = 0$ at $x = 0$ and $dy/dx = 0$ at $x = 0$. Substituting into Eq. 1 and its derivative yields $C_1 = C_2 = 0$. The equation of the curve then becomes

$$y = \frac{w_0}{2F_H}x^2 \qquad (2)$$

EXAMPLE 7.12 CONTINUED

This is the equation of a *parabola*. The constant F_H may be obtained using the boundary condition $y = h$ at $x = L/2$. Thus,

$$F_H = \frac{w_0 L^2}{8h} \tag{3}$$

Therefore, Eq. 2 becomes

$$y = \frac{4h}{L^2} x^2 \tag{4}$$

Since F_H is known, the tension in the cable may now be determined using Eq. 7–10, written as $T = F_H / \cos\theta$. For $0 \leq \theta < \pi/2$, the maximum tension will occur when θ is *maximum*, i.e., at point B, Fig. 7–21*a*. From Eq. 2, the slope at this point is

$$\left.\frac{dy}{dx}\right|_{x=L/2} = \tan\theta_{\max} = \left.\frac{w_0}{F_H}x\right|_{x=L/2}$$

or

$$\theta_{\max} = \tan^{-1}\left(\frac{w_0 L}{2F_H}\right) \tag{5}$$

Therefore,

$$T_{\max} = \frac{F_H}{\cos(\theta_{\max})} \tag{6}$$

EXAMPLE 7.12 CONTINUED

Using the triangular relationship shown in Fig. 7–21*b*, which is based on Eq. 5, Eq. 6 may be written as

$$T_{\max} = \frac{\sqrt{4F_H^2 + w_0^2 L^2}}{2}$$

Substituting Eq. 3 into the above equation yields

$$T_{\max} = \frac{w_0 L}{2}\sqrt{1 + \left(\frac{L}{4h}\right)^2} \qquad \textit{Ans.}$$

For a differential segment of cable length ds, we can write

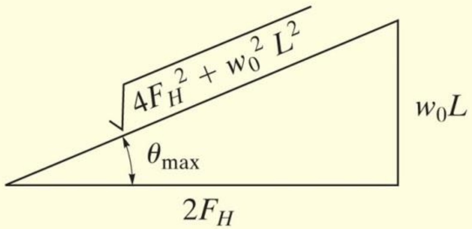
$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Hence, the total length of the cable can be determined by integration. Using Eq. 4, we have

$$\mathscr{L} = \int ds = 2\int_0^{L/2} \sqrt{1 + \left(\frac{8h}{L^2}x\right)^2}\, dx \qquad (7)$$

Integrating yields

$$\mathscr{L} = \frac{L}{2}\left[\sqrt{1 + \left(\frac{4h}{L}\right)^2} + \frac{L}{4h}\sinh^{-1}\left(\frac{4h}{L}\right)\right] \qquad \textit{Ans.}$$



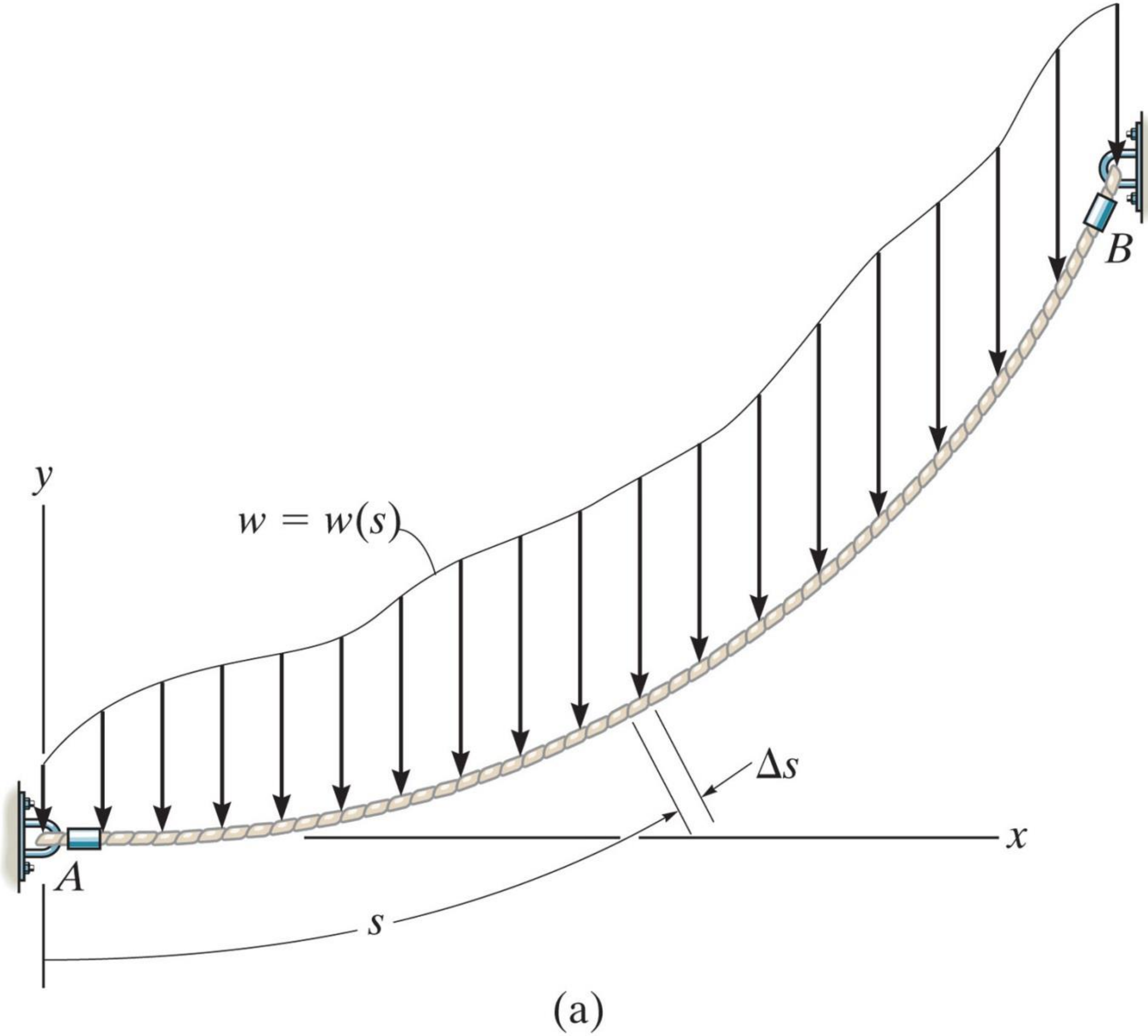
(b)

Fig. 7–21 (cont.)

7.4 Cables

Cable Subjected to its Own Weight

- When the weight of the cable is considered, the loading function becomes a function of the arc length s rather than length x
- Consider a loading function w=w(s)



(a)

Figure: 07_022a

s

n Weight

e is considered,
s a function of the
th x

w=w(s)

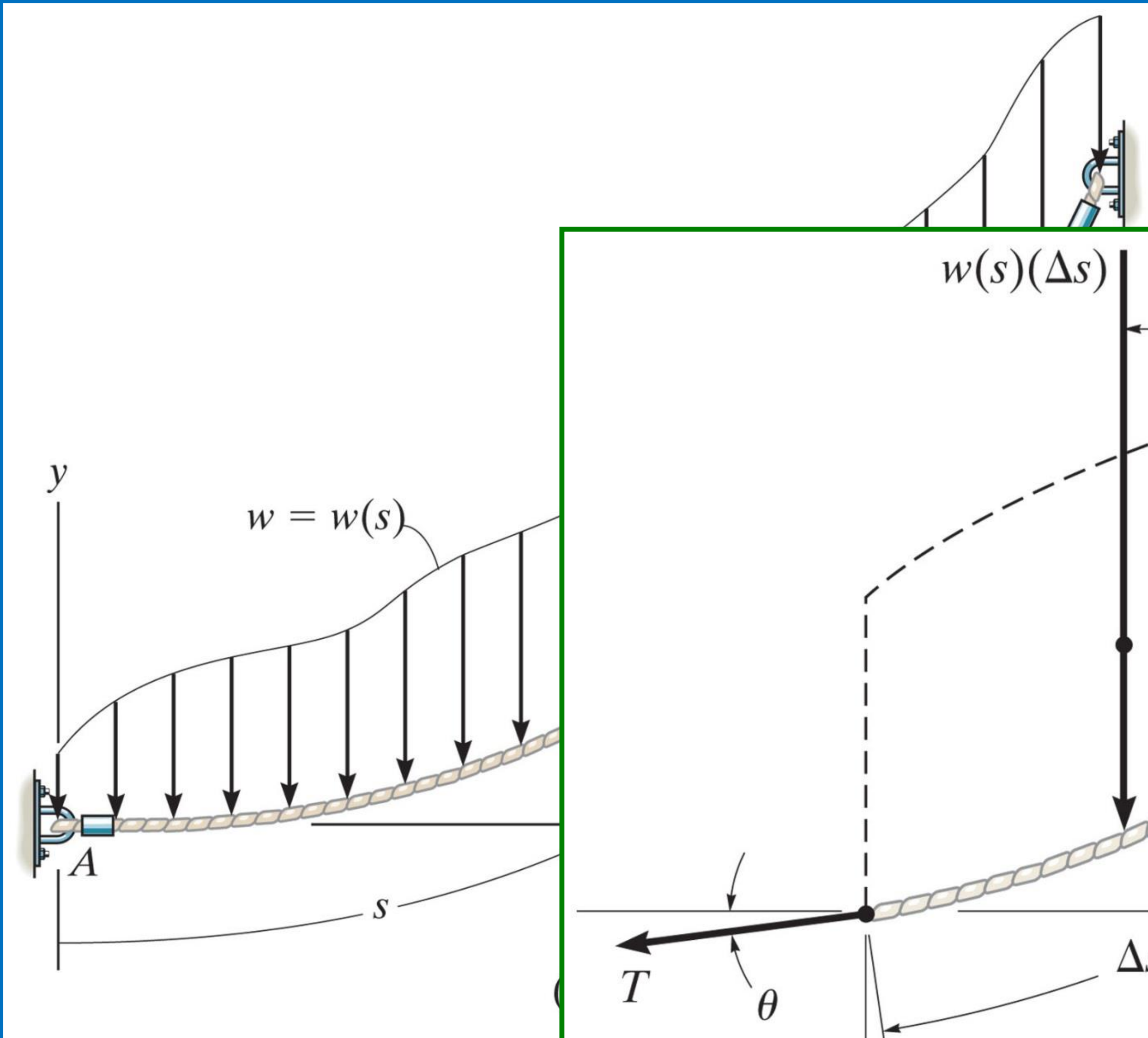
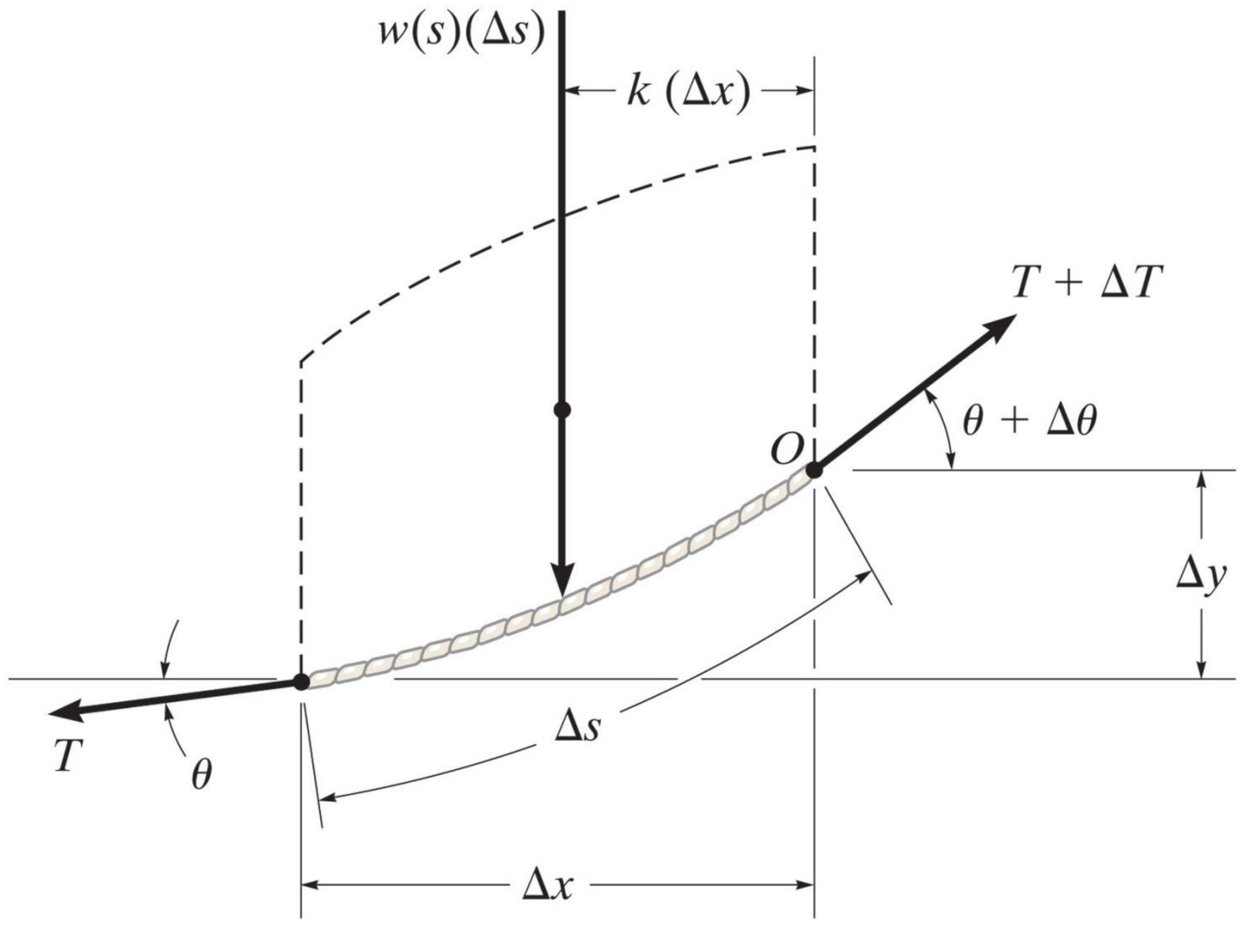


Figure:



(b)

Figure: 07_022b

7.4 Cables

Cable Subjected to its Own Weight

Apply equilibrium equations to the force system

$$\boxed{\begin{aligned} T\cos\theta &= F_H \\ T\sin\theta &= \int w(s)\,ds \\ \frac{dy}{dx} &= \frac{1}{F_H}\int w(s)\,ds \end{aligned}}$$

Replace dy/dx by ds/dx for direct integration

Since $ds = \sqrt{dx^2 + dy^2}$ Then $\frac{dy}{dx} = \sqrt{\left(\frac{ds}{dx}\right)^2 - 1}$

7.4 Cables

Cable Subjected to its Own Weight

Therefore

$$\frac{ds}{dx} = \left[1 + \frac{1}{F_H^2}\left(\int w(s)ds\right)^2\right]^{1/2}$$

Separating variables and integrating

$$x = \int \frac{ds}{\left[1 + \frac{1}{F_H^2}\left(\int w(s)ds\right)^2\right]^{1/2}}$$

7.4 Cables

Figure: 07_PH010

Electrical transmission towers must be designed to support the weights of the suspended power lines. The weight and length of the cables can be determined since they each form a catenary curve.

EXAMPLE 7.13

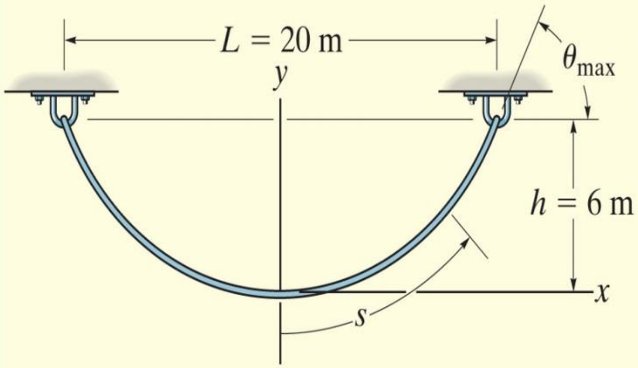


Fig. 7–23

Determine the deflection curve, the length, and the maximum tension in the uniform cable shown in Fig. 7–23. The cable has a weight per unit length of $w_0 = 5\ \text{N/m}$.

SOLUTION

For reasons of symmetry, the origin of coordinates is located at the center of the cable. The deflection curve is expressed as $y = f(x)$. We can determine it by first applying Eq. 7–15, where $w(s) = w_0$.

$$x = \int \frac{ds}{\left[1 + (1/F_H^2)\left(\int w_0\, ds\right)^2\right]^{1/2}}$$

Integrating the term under the integral sign in the denominator, we have

$$x = \int \frac{ds}{[1 + (1/F_H^2)(w_0 s + C_1)^2]^{1/2}}$$

EXAMPLE 7.13 CONTINUED

Substituting $u = (1/F_H)(w_0 s + C_1)$ so that $du = (w_0/F_H)\,ds$, a second integration yields

$$x = \frac{F_H}{w_0}(\sinh^{-1} u + C_2)$$

or

$$x = \frac{F_H}{w_0}\left\{\sinh^{-1}\left[\frac{1}{F_H}(w_0 s + C_1)\right] + C_2\right\} \qquad (1)$$

To evaluate the constants note that, from Eq. 7–14,

$$\frac{dy}{dx} = \frac{1}{F_H}\int w_0\,ds \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{F_H}(w_0 s + C_1)$$

EXAMPLE 7.13 CONTINUED

Since $dy/dx = 0$ at $s = 0$, then $C_1 = 0$. Thus,

$$\frac{dy}{dx} = \frac{w_0 s}{F_H} \tag{2}$$

The constant C_2 may be evaluated by using the condition $s = 0$ at $x = 0$ in Eq. 1, in which case $C_2 = 0$. To obtain the deflection curve, solve for s in Eq. 1, which yields

$$s = \frac{F_H}{w_0}\sinh\left(\frac{w_0}{F_H}x\right) \tag{3}$$

Now substitute into Eq. 2, in which case

$$\frac{dy}{dx} = \sinh\left(\frac{w_0}{F_H}x\right)$$

EXAMPLE 7.13 CONTINUED

Hence,

$$y = \frac{F_H}{w_0}\cosh\left(\frac{w_0}{F_H}x\right) + C_3$$

If the boundary condition $y = 0$ at $x = 0$ is applied, the constant $C_3 = -F_H/w_0$, and therefore the deflection curve becomes

$$y = \frac{F_H}{w_0}\left[\cosh\left(\frac{w_0}{F_H}x\right) - 1\right] \qquad (4)$$

This equation defines the shape of a ***catenary curve***. The constant F_H is obtained by using the boundary condition that $y = h$ at $x = L/2$, in which case

$$h = \frac{F_H}{w_0}\left[\cosh\left(\frac{w_0 L}{2F_H}\right) - 1\right] \qquad (5)$$

Since $w_0 = 5\text{ N/m}$, $h = 6\text{ m}$, and $L = 20\text{ m}$, Eqs. 4 and 5 become

$$y = \frac{F_H}{5\text{ N/m}}\left[\cosh\left(\frac{5\text{ N/m}}{F_H}x\right) - 1\right] \qquad (6)$$

$$6\text{ m} = \frac{F_H}{5\text{ N/m}}\left[\cosh\left(\frac{50\text{ N}}{F_H}\right) - 1\right] \qquad (7)$$

EXAMPLE 7.13 CONTINUED

Equation 7 can be solved for F_H by using a trial-and-error procedure. The result is

$$F_H = 45.9 \text{ N}$$

and therefore the deflection curve, Eq. 6, becomes

$$y = 9.19[\cosh(0.109x) - 1] \text{ m} \qquad \textit{Ans.}$$

Using Eq. 3, with $x = 10$ m, the half-length of the cable is

$$\frac{\mathscr{L}}{2} = \frac{45.9 \text{ N}}{5 \text{ N/m}} \sinh\left[\frac{5 \text{ N/m}}{45.9 \text{ N}}(10 \text{ m})\right] = 12.1 \text{ m}$$

Hence,

$$\mathscr{L} = 24.2 \text{ m} \qquad \textit{Ans.}$$

Since $T = F_H/\cos\theta$, the maximum tension occurs when θ is maximum, i.e., at $s = \mathscr{L}/2 = 12.1$ m. Using Eq. 2 yields

$$\left.\frac{dy}{dx}\right|_{s=12.1 \text{ m}} = \tan\theta_{\max} = \frac{5 \text{ N/m}(12.1 \text{ m})}{45.9 \text{ N}} = 1.32$$

$$\theta_{\max} = 52.8^\circ$$

And so,

$$T_{\max} = \frac{F_H}{\cos\theta_{\max}} = \frac{45.9 \text{ N}}{\cos 52.8^\circ} = 75.9 \text{ N} \qquad \textit{Ans.}$$